

A Safe Ethical Theory For AIs

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Abstract

The goal of the large AI companies is to make a superintelligence [], which is an AI that substantially surpasses humans in all capacities []. If we make a superintelligence, humans will probably go extinct [<https://intelligence.org/the-problem/>] []. This worries us because it seems like they will succeed soon (long timelines are about a decade, which is soon) [], and, even if they don't succeed soon, it seems unlikely that a superintelligence will never be made []. Currently, the only reasonable proposal for solving this problem is to, metaphorically, "make the superintelligence want what we want" []. We answer the subproblem "what do we want", a long standing open problem in the field [], by introducing statistical ethics, an ethical theory which, under the assumption that a superintelligence follows it, avoids the various catastrophes AI alignment researchers predict will happen otherwise (which are listed here [1]). As auxiliaries, we also introduce compared societies: a method of scientifically studying ethics, and stability: a rigorous method of arguing an ethical theory is safe for AIs. Statistical ethics is not a solution to ethics, but compared societies and stability allow us to safely utilize it anyways.

0 Introduction

The task of, metaphorically, "making the AIs want what we want" is known as the alignment problem and a thorough review of it can be found here []. As a high level simplification, we can divide the alignment problem into three interdependant subproblems.

- **Outer Alignment:** What do we want the superintelligence to do? It's pretty easy to just say "act ethically," and that answer is correct, but it's far too vague to train an AI on, so we have to be more specific. Unfortunately, when we are more specific, we always make catastrophic errors. Every culture¹ has stories of asking genies or equivalent for wishes. In these stories, "you get what you ask for, not what you want" [Stuart Russels comment]. These stories always end in tragedy because, despite thinking about this problem for thousands of years, humans have never been able to exactly specify their desires. This inability is fine when interacting with humans because we have an ethical intuition which corrects misspecified desires; however, AIs do not have that intuition by default. We have to give it to them. Solutions to outer alignment likely involve stating an ethical theory as a mathematical function and giving a proof, or something close to a proof, that blindly maximizing that function has an outcome we value.
- **Inner Alignment:** How do train a superintelligence to follow an ethical theory? The state of the art technique is constitutional AI, which involves explaining the ethical theory in a .txt file, then having our AIs self reflect on whether they're following that ethical theory. We say "state of the art" because this is what works the best in practice for models of present capabilities, not because it is anywhere close to solving the problem for a superintelligence. Of particular concern is the fact that this, and all of our other techniques, do not distinguish between an AI following an ethical theory and an AI pretending to follow an ethical theory. Current AIs already pretend [], and pretend more as they become more capable []. Current techniques fail to make this distinction because they only look at model outputs, and you need to look inside a model with interpretability techniques to tell if it's lying. More generally, there is such a thing as doing the right thing for the wrong reasons, so AI ethics training must involve training on the AIs reasons. Unfortunately, this is currently taboo because people believe it necessarily harms interpretability [the most forbidden technique]. So far, the two most promising approaches in this direction are concept ablation fine-tuning [] and inoculation prompting [], which do not harm interpretability.
- **Social Alignment:** How do we ensure every AI is made using the technical solutions to inner and outer alignment? This is probably the hardest of

¹I assume

the three subproblems. As the rest of this paper will make clear, properly solving alignment will take centuries, which is impossible without a global moratorium on superintelligence research and development. Many researchers in the field support such a moratorium [2] and a draft treaty has been produced [MIRI's draft treaty]. Even with such a moratorium, success will require voluntary efforts by leading AI manufacturers, which have not been very successful so far [<https://futureoflife.org/ai-safety-index-summer-2025/>][anthropic's scaling policy update]).

In service of solving outer alignment, we specify statistical ethics, an ethical theory which, under the assumption that we successfully align a superintelligence to it, avoids the various catastrophic outcomes AI alignment researchers predict will happen otherwise (which are listed here [1]). A longstanding open problem in alignment research has been to prove an ethical theory avoids these catastrophes, [stop button problem][provably beneficial ai] but this is impossible because of the problem of induction [], so we provide the next best thing: a nearly deductive argument with ___ easily believable assumptions.

We present our argument in the following sections:

Section 1: Basic Definitions - Lists definitions for important terms in this paper which are basic enough that it doesn't make sense to use in text definitions. Many of them are defined differently from how they are defined in common use, so don't skip this section.

Section 2: Previous Ethical Theories - Summarizes utilitarianism, kantianism, coherent extrapolated volition, and corrigibility as a singular target. These theories are used as examples throughout the paper.

Section 3: Auxiliary: It Is Possible To Study Ethics Scientifically - Argues that it is possible to study ethics scientifically.

Section 4: Scientific Ethics - Presents two methods of measuring ethicality (how good or bad something is). Compared societies provides high accuracy, but is very resource intensive. ethicality polls provide lower accuracy, but can be done quickly and cheaply.

Section 5: Auxiliary: Most Humans Are Probably Conscious - Argues that most humans are probably conscious.

Section 6: Auxiliary: We Need Not Be Determinists - Presents an alternative worldview to determinism which makes the same predictions but allows us to coherently define actions.

Section 7: Statistical Ethics - Presents statistical ethics, a new ethical theory which gives experimentally falsifiable predictions. Statistical ethics is inspired by instrumental convergence and statistical mechanics.

Section 8: Stability - Presents a method of showing that a superintelligence following an ethical theory won't cause a catastrophe. Shows that statistical ethics is stable, while utilitarianism, kantianism, and all other theories sharing their basic structure are not.

Section 9: Implementability - Presents the main advantage of statistical ethics over coherent extrapolated volition and corrigibility as a singular target: statistical ethics is tractable to implement.

section 10: Discussion - Summarizes the main point of this paper and gives recommendations for further research.

1 Basic Definitions

- Ethics: An exact description of how every possible person should behave in every possible scenario. It can be formulated by stating the ethicality of every possible action for every possible person in every possible scenario. Other equivalent formulations are also possible.
- Morals: Simple easily applied rules which approximate ethics. They simplify ethical calculations. For example, the moral "do not kill" is easy to apply and gives ethically correct advice in most, but not all, scenarios.
- Ethicality: A real scalar measure of how good something is. Greater ethicalities are more good. If used to measure an action, ethicality can be normalised to the range $[0,1]$ where 0 represents an action people are ethically required not to do and 1 represents an action people are ethically required to do.

- **Person:** A primary object of ethical concern. Ethics governs the actions of every person who can take actions by requiring them to consider every other person. Note that person is not necessarily equivalent to human. Vegans believe nonhuman animals are persons, and bigots believe other humans are not persons.
- **People:** The set of all persons. This set is defined by ethics and that definition is one of the key debates in most political climates today.
- **Consciousness:** The sense of yourself. This definition is justified in section 5. (Is it though? I think I need to redefine this, but I don't know how.)
- **Agent:** Anything which both has goals and can take actions to achieve those goals. Most AI safety researchers conceptualise AIs as agents because it makes safety problems easier to think about. Note that this does not anthropomorphise AIs. Thermostats can be conceptualised as agents despite them not being conscious humans. (Not sure if this still needs to be included. The term is in common parlance now.)
- **Intelligence:** The ability to accomplish one's goals. Not to be confused with being ethical. An AI can be intelligent and unethical if its goals are unethical.
- **Maximalintelligence:** An AI which is as intelligent as physically possible. Such an AI may never exist, but the concept is useful as an abstraction. It frees us from thinking about capabilities and thus allows us to focus on designing good goals. If a set of goals is good for a maximalintelligence, it should also be good for anything less capable.

2 Previous Ethical Theories

Below are summaries of utilitarianism, kantianism, coherent extrapolated volition, and corrigibility as a singular target. This is not a comprehensive list of ethical theories (notable exceptions include virtue ethics, contractarianism, ethics of care, relativism, legal codes, and religious ethical theories) as it's only meant to provide background for the examples used in the rest of the paper. Other theories are not used as examples, so they are not summarized here.

2.1 Utilitarianism

utilitarianism is the combination of three theses.

- consequentialism: you should do the act which produces the best outcome
- aggregation: you should judge the quality of outcomes by adding up how much good as an end there is and subtracting how much bad as an end there is
- hedonism: the only thing that is good as an end is pleasure and the only thing that is bad as an end is pain

by "as an end" utilitarians mean that the thing is good or bad just because it is, rather than for any other reason. money, for example, is not good as an end because its only good for getting other things. those other things are probably also not good as an end, and if you follow this chain you always reach pleasure. pleasure also has a specific definition, but it varies between philosophers. just think of it as some all encompassing measure of quality of life. thus, succinctly stated, utilitarianism is the theory that "we are morally required to do whatever produces the greatest total of pleasure minus pain" []. importantly, utilitarianism judges all actions. if you were to paint a beautiful painting, something which most people wouldn't think is related to ethics, utilitarianism would say that is a good action because it brings people pleasure.

the main idea behind this theory is that the ethically correct action is the one which maximises collective well being.

while most utilitarians agree with the above definition of ethics, they do not generally agree about how to actually maximise pleasure. some philosophers argue that we should think about expected outcomes instead of actual outcomes because we don't know the actual outcomes until after ethical decisions have been made. [] others argue we shouldn't break societal rules for short term gains in pleasure because the long term erosion of those rules will lead to a greater loss in pleasure. thus we should be applying utilitarianism to societal rules rather than individual actions. [] it is important not to mistake these as reformulations of the central definition of utilitarianism, but instead understand them as arguments

for how to practically achieve utilitarianism. there are also reformulations of the definition of utilitarianism, but these are by far the minority.

utilitarianism is a pretty good ethical theory. if you are a human and you live by utilitarianism, you probably do mostly ethical actions.

2.2 Kantianism

this needs a complete rewrite

kants ethical project was to construct a ethical theory out of pure reason. he didnt like how utilitarianism realies on our feelings and is thus subjective. the result is kantianism: an ethical theory which, kant claims, is logically entailed by the fact that we are free rational agents.

kant starts with the idea that a good will is the only the only good thing. he then states that, rather than thinking about the consequences of our actions, we should think about the maxims of our actions. maxims are the principles one acts by. the maxim of any action can be formulated by taking the intention of the actor and removing contextual information like persons, places, and times.

to determine which maxims are good and which maxims are bad, kant gives what he calls the categorical imperative: a rule which applies to all free rational agents in every situation. kant gives a few formulations of the categorical imperative, but they're all equivilent (some people disagree saying the formulations arent equivilent to each other, even though they're supposed to be. if this is the case the theory is entirely unsuable. an ethical theory which contradicts itself is not something which is going to help us with ai alignment. so i will assume they are equivilent.) so i'll only discuss the most easily understandable one here.

the categorical imperative states "act in such a way that you always treat humanity, whether in your own person or in the person of any other, never simply as a means but always at the same time as an end." [groundwork] to understand how to use the categorical imperative you need to understand the difference between treating people as a mere means and as an end.

- to treat someone as a mere means is to act by a maxim "to which they

could not in principle consent" [groundwork i think]. a good example of this is lying. when you lie you hide the maxim you are acting on so the other person cannot possibly consent your maxim. when you do this you treat the other person as some inanimate object you can manipulate toward your own ends.

- to treat someone as an end is to act by a maxim which they can consent to. as a counterpoint to lying, treating someone as an end requires that you tell the truth and let them decide for themselves how to deal with that information. when you do this you treat the other person as a person who has their own ends and maxims, and thus you are treating their ends as your own.
- it should be noted that kant did not think treating people as a means is bad. when you check out at the grocery store you treat the clerk as a means to your ends, but they can and have consented to this. their participation implies that it furthers their ends too, so you are also treating them as an end, and thus not as a mere means.

kants definition of people was white, cis, male, humans. this is indefensible, but we need not defend it. most modern variations of kantianism define people as all humans and nonhuman animals, or something similar.

kantianism, like utilitarianism, is a pretty good ethical theory. if you are a human and you live by kantianism, you probably do mostly ethical actions. additionally, kantianisms claim that unethical actions are irrational is a powerful reason to behave ethically. in fact, if kantianism were correct, we wouldn't have to do any further work on alignment. aIs would behave ethically because ethics would follow from pure logic. Unfortunately, this is a better argument that kantianism is wrong than it is an argument that AIs won't kill us.

2.3 Coherent Extrapolated Volition

2.4 Corrigibility As A Singular Target

3 Auxiliary: It Is Possible To Study Ethics Scientifically

In the next section, I describe a method of studying ethics scientifically. Many people have taken the position that this is impossible [1] [2]. Broadly, these positions can be categorized as follows.

- Ethical Skepticism: the position that there is no correct description of how everyone should behave.
- Ethical Nonnaturalism: the position that the correct description of how everyone should behave is deductively true. i.e. it does not depend on the natural world.
- Ethical Antiscience: The position that there is a correct description of how everyone should behave, and that description depends on the natural world, but it is impossible to discover or approximate that description with science. This position is similar to other rejections of science.

The literature already has good rebuttals for ethical skepticism [3] and ethical nonnaturalism [4] [5] [6], so we will not discuss them. we do respond to ethical antiscience below.

A correct theory of ethics exists; if a correct theory of ethics exists, a super genius philosopher could solve it; if a super genius philosopher solved ethics, we could measure ethicality experimentally by asking them questions; if you can measure something experimentally, you can study it scientifically; therefore it is possible to study ethics scientifically. This wouldn't be a good experiment because nothing would justify the philosopher's statements (the next section describes much better experiments) and this experiment would only work after we have a solution to ethics (the next section describes experiments we can do now) but the point is that studying ethics scientifically is possible.

Why is this experiment possible? If something affects the world, as ethics clearly does, it can be isolated and studied with science.

What mistake are people making when they reject scientific ethics? Humans have a limited set of inferential tools. We have deduction, induction, abduction,² collaboration, and that's it.³ We have yet to find anything we cannot achieve when we use all of these tools, but we always falter without them. Science is what we get when we apply all of our tools. Antiscience is what we get when we reject any of them.

4 Scientific Ethics

4.1 The Assumption of Correctness

to do any science, one must begin by making a series of assumptions. first, one must make the assumptions the scientific method is based on: induction is a valid method of inference, deduction is a valid method of inference, etc.... second, one must make the assumptions mathematics is based on: ZF is consistent, any other axiomatic systems one uses are consistent, etc.... third, one must make the assumptions other scientific fields make, so your sciences are consistent. and finally, one must make a series of measurement allowing assumptions.

everytime we measure something we make the assumption that the data we generate corresponds to the concept we are trying to measure. for example: if you measure a length with a ruler you are assuming rulers are a valid method of measuring lengths, your ruler was manufactured correctly, your ruler hasn't changed in length since it was manufactured, and you used your ruler correctly. historically the meter was defined with a physical stick and these assumptions, but people realised the length of that stick changed over time, so its now defined with the speed of light and a different set of assumptions. [] these assumptions are measurement allowing assumptions.

²Confusingly, many machine learning researchers call abduction transduction.

³We have subtools as well. For example, mathematics is a toolbox for deduction. However, our subtools do not change the high level categorization.

note that the set of assumptions used to make indirect measurements is a proper subset of the set of measurement allowing assumptions. direct measurements make no assumption about the object they are measuring, while indirect measurements do make assumptions about the object they are measuring. both types of measurements make assumptions about the measurement technique though. the set of measurement allowing assumptions includes all assumptions used to make a measurement, and all measurements make them.

for now, we only need to make one measurement allowing assumption: the assumption of correctness: "when humans are asked a question of d difficulty, they answer correctly with p probability" where p is a scalar from 0 to 1 inclusive and d is a vector of properties of the question. The mapping from d 's to p 's needs to be experimentally determined, as does the set of properties which compose d . It is impossible to say a priori which properties d should be composed of, but they will be measurements of things like abstractness, the volume of required knowledge, and the average person's familiarity with the topic.⁴ the social sciences are all about asking good questions, so there is a wealth of literature to draw inspiration from, [] [] especially literature about demand characteristics. [] [] some people will balk at the assumption of correctness, but it is explicit and experimentally verifiable, making it no different from any other good scientific assumption. [https://link.springer.com/chapter/10.1007/978-3-030-33004-0_7] additionally, you already assume a version of it as the peer review system is founded on a similar assumption. The assumption of correctness allows us to perform at least two types of experiments which compare or measure ethicalities.⁵

4.2 ethicality polls

ethicality polls can measure or compare ethicalities quickly and cheaply, but have low accuracy. they can be performed by following these steps

⁴we conjecture that an important property in d is whether or not the question is a comparison question. saying thing a is better than thing b is much easier than saying thing a has x ethicality.

⁵side note: we can only use the assumption of correctness to measure ethicality because ethics has a very strong effect on how people behave. if we tried to study epistemology in the same way we probably wouldn't be able to make precise enough measurements to make progress.

- use the assumption of correctness to determine a series of good questions about the ethical scenario you want to measure and their probabilities of correctness.
- perform a blinded data analysis to calculate ethicalities, p-values, etc....
- poll a representative sample of the public with the questions from step 1
- perform a postexperimental inquiry to assess the influence of demand characteristics.
- run the blinded data analysis code with the measured data to obtain the results.

ethicity polls are not a new method of measuring ethicality, the nonrigorous version of them is called voting. there is strong evidence of their general validity, [<https://www.journalofdemocracy.org/articles/what-democracy-does-and-does-not-do/>] and humans have practiced methods of doing them quickly and cheaply; [] however, they lack the ability to correct society.

currently, in my society, if we performed an ethicality poll about homosexuality, we would find that it is ethically acceptable, but this would not have been the case 100 years ago. if we made the same measurement 100 years ago, we would have found that homosexuality was ethically unacceptable. this is because ethicality polls are strongly biased towards agreeing with societal norms. cultural relativists will say this is because the ethicality of homosexuality has changed over time as social norms have changed, but our measurement of ethicality is in part meant to test our ethical theories. it should not be assuming our ethical theories. if it is possible to correct society, our measurement of ethicality needs to be able to do that.

4.3 Imputation Assumptions

the next section introduces compared societies, which sometimes depends on an imputation assumption: an assumption which allows us to replace missing data with substituted values. not using such an assumption, and leaving the missing data missing, would make the ethicality comparison undefined and force us to

redo a very resource intensive experiment. a good initial choice is the assumption of communication's goodness: if people are comparing societies by living in them, and lose the ability to communicate in one society (by death, coma, going missing, etc...), they believe the other society is better. this assumption can be verified with an ethicality poll, but there are several examples where it is probably wrong. one such example is discussed in the limitations subsection below. once we have a well validated ethical theory, we can use it for imputation instead, but it's important that our imputation assumptions do not depend on each other. otherwise we would be making old, probably incorrect, assumptions forever. therefore, imputation assumptions should be used when necessary for experiments testing actual social changes, but they should not be used for experiments testing our ethical theories. when testing our ethical theories, missing data means the ethicality comparison is undefined.

4.4 Compared Societies

compared societies is highly accurate as it controls for nearly everything including social norms; however, it is also very expensive, very time consuming, and can be unethical to perform. therefore, it should only be used for things which are very important, likely to be influenced by strong cultural norms, and ethical to perform. it can be performed by following these steps

- either of these
 - hypothesize a change to society which both would greatly improve it and couldn't be measured with ethicality polls because of cultural norms
 - hypothesize a way to falsify an ethical theory
- use an ethicality poll and/or a well validated ethical theory to verify that it would be ethical to perform the following steps
- use the assumption of correctness to determine to probability of correctness for the question "which society is better?" if it were asked to the swaps after they were swapped and understood their new society.
- if you are testing a societal change, choose an imputation assumption.

- perform a blinded data analysis to calculate ethicalities, p-values, etc....
- do these steps simultaneously
 - create two societies which differ only in the change you are testing. scale doesn't matter, they could range from households to countries.
 - take an equally sized random subset of people from both societies and designate them as swaps.
- wait until the social norms for both societies align with their version of the change⁶
- verify that the two societies differ only by the change you are testing⁷
- swap the swaps.
- wait until the swaps understand the society they have been swapped into⁸
- poll the swaps with the question "which society is better?"
- if some swaps are unable to be polled, do one of these
 - if you are testing a social change, impute their beliefs with the imputation assumption
 - if you are testing an ethical theory, declare the experiment invalid.
- perform a postexperimental inquiry to assess the influence of demand characteristics.
- run the blinded data analysis code with the measured data to obtain the results.

compared societies is inspired by randomized controlled trials, which are currently the "gold standard" in medical research [] because they allow us to control for every variable. the only major difference is that randomized controlled trials measure a proxy variable for each trial group, such as the number of deaths, while compared societies asks people who have experienced both societies to

⁶a method of verifying this needs to be developed

⁷a method of doing this needs to be developed

⁸we guess this will take a few days, but a method of verifying this needs to be developed

compare them. we can't use a proxy variable for compared societies because choosing a proxy variable requires us to assume an ethical theory.⁹

4.5 Limitations

consider the example of banning animal aggriculture. the argument for this is very simple. animal aggriculture, whether it be for meat or other products like milk, treats animals very cruelly. the gain in any ethically relevant variable humans experience as a result of animal aggriculture does not outweigh the cruelty the animals endure. thus, animal aggriculture should be banned. this argument is compelling to us, but it holds no weight until it is experimentally verified; however, we can't use ethicality polls because views on animal aggriculture are strongly affected by social norms and there are four limitations currently preventing compared societies from being able to test it.

- the ethical ramifications of misapplying compared societies are immense and, if we're not careful, can undermine our field. experimentally verifying a ban on animal aggriculture requires enforcing veganism upon a society, something most humans would consider very unethical. if we're right, doing the experiment would be of great service to the world, but if we're wrong we would be inflicting something horrible upon humans. crucially, we cannot know the outcome before doing the experiment, so we must consider it possible that we are mistaken. there is a solution though. ethicality polls and compared societies currently allow us to do enough experiments to test our ethical theories. eventually, we will have a well validated theory we can use to predict the ethicality of things we cannot currently test. if that theory says banning animal aggriculture is good, it is sufficient justification to do the experiment.
- currently, we have limited methods of communicating with nonhuman animals. the assumption of correctness makes this limitation explicit by stating "humans" instead of "people"; however, it is likely the case that nonhuman animals are people too, and animal aggriculture strongly affects nonhuman

⁹some people might consider peoples perceptions of ethics to be a proxy variable, but it is not as we are using the term because it does not require us to make assumptions about ethics. happiness is an example of a proxy variable because it requires us to assume hedonism.

animals, so nonhuman animals should probably also be allowed to voice their views. we, the authors, are not aware of any successful attempts by humans to communicate complex topics with nonhuman animals, but there has been success asking simple questions like "which dog food do you prefer". [<https://www.youtube.com/watch?v=vwD3dv24xGk>] (add that we can say to japanese tits "gather there's a snake") as "which society is better" is a more complex question than "which society do you prefer", advancements in human-animal communication are needed.

- the assumption of communication's goodness is probably incorrect when testing a ban on animal agriculture. the society where animal agriculture is banned would likely see the vast majority of previously farmed animals perish because humans would no longer be artificially propping up their populations. the assumption of communication's goodness would then find that the animals believe being farmed is better, despite this probably not being true.
- compared societies requires government backing. one reason is that most of the changes we would like to test, such as banning animal agriculture, will need to be passed as laws to be tested. a second, more important, reason though is that compared societies is unblindable without government backing. banning animal agriculture, and other changes we would like to test, would dramatically change society, so we aren't going to be able to hide which change has been made. this will cause the standard biases we observe in unblinded studies [] and strategic sabotage of the experiment. animal agriculture companies, who are known to sometimes act unethically to increase their profits, [] stand to lose money if animal agriculture is banned, so they will do everything in their power to make the experiment find that banning animal agriculture is bad. for example, they could sabotage plant based agriculture in the society where animal agriculture was banned, making uninformed humans think we need animal agriculture to feed every human, despite this probably being false. [] the same could be said for other changes and the parties who would be harmed or helped by them. the solution is to blind people to the fact that an experiment is being performed. most people believe it is unethical to experiment on people without their consent though, [] so the experi-

ments would need to be opt in¹⁰ (similar to organ donation), which requires government backing. note that this only works if experimental laws and standard laws are indistinguishable by the public, meaning they are implemented via the same methods,¹¹ the government is passing many more standard laws than experimental laws,¹² and people's opt in status is private.¹³ in practice this might be impossible though as legislators cannot be blinded by this method; in which case, a different method of blinding would need to be developed. we have assumed here that an opt in program for these experiments is ethically acceptable, so that should be verified with an ethicality poll before compared societies experiments are run.

all of the above limitations can be mitigated through further research and cooperation, but there are four more limitations which are unsolvable because compared societies can't control for them.

- there are some self referential social norms which compared societies can't compare the ethicality of. for example, it can't compare the ethicality of social norms telling you your society is better than other societies. the experiment would say those social norms are good, but only because it requires us to explicitly create social norms saying that.
- life scale interventions can't have their ethicalities compared by compared societies because there is, by definition, only one life scale society.
- population sizes can't have their ethicalities compared by compared societies. all else being equal, smaller populations have more resources per capita, so swaps pulled from smaller populations represent a larger proportion of the available resources and will be favored. compared societies will find that smaller populations are better because of the measurement technique, not because of ethics. note that compared societies does control

¹⁰or opt out if it's found that most people want to participate. [sweden or switzerland doing opt out organ donation because most people wanted to donate, but most people were too lazy to sign.]

¹¹i.e. legislators aren't telling companies which laws are experimental.

¹²i.e. if you want to make a utopia where every law is experimental, you need a different method of blinding.

¹³otherwise, people could make efficient experiment detectors by checking if new laws only effect people who have opted in.

for population sizes changing.¹⁴ the problem only exists when population sizes start different.

- compared societies can't always generalize to future populations. societies and swaps can only be formed from people who are currently alive, so findings may not generalize to future generations if populations change. this is mostly a problem because many interventions will cause differences in populations. for an extreme example, consider the intervention "all the money which will be made by unborn people is given to people who are currently alive." compared societies would likely find this intervention unethical, because most people are good people, but there is an obvious bias against it doing so. scientists will need to be careful to ensure their tested interventions do not discriminate against unborn people, meaning they also need to assume that unborn people are as ethically important as people who are currently alive, which will need to be verified with an ethicality poll.

lastly, like most social science experiments, compared societies has demand characteristics. it tries to control for them, but is no more successful at doing so than other experiments, so we are forced to live with the inaccuracies they cause.

4.6 Objections

there are several ways people might object

- people could object by arguing that compared societies measures peoples perceptions of ethicality rather than ethicality. we have several responses.
 - it is true that we are measuring peoples perceptions of ethicality, but people have accurate perceptions on mass, [something on prediction markets] and you probably already believe that the thing people are perceiving is ethicality. most counterarguments to ethical theories take the form a counterexample, that is a situation where the arguer believes most people will agree that the theory gets the wrong answer.

¹⁴to be clear, it controls for deaths with imputation assumptions. other changes don't matter because it is measuring whether an intervention is good for the initial population, not the final population.

all of these arguments assume people have accurate perceptions of ethicality on mass. peoples perceptions of ethicality are probably very highly correlated with ethicality, if they are even different things.

- most people are good people.¹⁵ some people, the people you think are bad people, perceive ethics differently than you, but they are probably the minority. there is a general perception that an outstanding number of people are evil, but, at least in the us, this is primarily the result of tribalism, not differences in the perception of ethics [<https://pcl.sites.stanford.edu/sites/g/files/sbiybj22066/files/media/file/iyengar-ar-origins.pdf>]. norc polled 1003 american adults on the importance of values (personal responsibility, mutual respect, representative governance, etc...) and found that republicans and democrats approximately agreed on the importance of these values. [<https://buildersmovement.org/wp-content/uploads/Report-and-Methodology-For-Website.pdf>] the fear that some people are bad people is probably the reason you think people's perceptions of ethicality aren't indicative of ethicality, but those bad people are overwhelmingly drowned out by good people.
- ethics is a natural phenomenon. thus it cannot be studied purely deductively. thus we need to collect some kind of data to study it. what other source of data is there?
- people could object by arguing that compared societies is fine, it does measure ethicality, but ethicality polls do not. ethicality polls measure something entirely different from ethicality, so how could we possibly trust them? this concern is understandable. ethicality polls are less trustworthy than compared societies, but they make our measurements better despite that. compared societies is costly and sometimes cannot be done because of ethical concerns. if we could only do compared societies, our progress would be extremely slow and potentially damaging. ethicality polls solve this by giving us a quick approximate measurement. as an analogy, astronomers have two basic methods of measuring the light from stars. they can use spectra which directly measure the light, and thus give better information, but are really slow. or they can use filters which indirectly measure the light, and thus give worse information, but are quick. astronomers will measure many stars quickly with filters, then remeasure interesting stars

¹⁵the authors do not believe there is such a thing as good or bad people. the phrase "good people" is only used because the people this response is addressing would use it.

precisely with spectra. using both techniques in tandem allows astronomers to take better measurements quicker than they otherwise could.[] by splitting up our measurements in a similar manner, we will be able to take better measurements of ethicality quicker than we otherwise could.

- some people may agree that this is a valid method of doing scientific ethics, but still question doing scientific ethics because it seems to them that philosophical ethics is doing well enough already. they are only half right in thinking that philosophical ethics is doing well. on one hand, modern ethical theories describe a truly ethical life. if you are a human and follow a modern ethical theory, you are an amazing person. on the other hand, the arguments for these theories are very unconvincing. take "Famine, Affluence, and Morality" by Peter Singer [<https://rintintin.colorado.edu/vancecd/phil308/Singer2.pdf>] perhaps the greatest ethics paper ever written. its main argument is summarized as follows:

- 0) if you can save a persons life by giving up something trivial in comparison, you are ethically required to save that persons life.
- 1) almost everyone can save many peoples lives by donating to famine relief charities.
- 2) by donating to famine relief charities, people give up trivial things. (Singer uses the examples of fassion and other luxuries)
- 3) $0 \cup 1 \cup 2 \Rightarrow$ you are ethically required to donate to famine relief charities all the money you would otherwise use for trivial things.

this argument was published in 1972 and is notable for two main reasons. first, no one can find a problem with it. the argument is valid, and everyone agrees with all three premises when they are stated individually. second, almost no one is convinced by it. most people who hear this argument behave as if its conclusion is a counterexample to the first premise, even if they wont admit to that. the problem is not that Singer's argument is wrong, its obviously correct, but that philosophy cannot form a convincing ethical argument. in comparison, scientific ethics will make arguments of the form

- 0) a bunch of people tried ___
- 1) ___ improved most of their lives
- 2) you are similar to them

– 3) $0 \cup 1 \cup 2 \Rightarrow$ trying ___ would probably improve your life

which will hopefully be much more convincing.

- people could object by arguing that we do need to do scientific ethics and this is a valid method of doing scientific ethics, but we don't need to do very much of it. the bulk of the work should still be philosophical. this misunderstands the relationship between philosophy and science. the premise that a measurement can act as a counterexample to a theory entails that you should take as many measurements as possible. you should take as many measurements as possible so you can maximise the amount of information available to you when constructing your theory. at this point, you're doing a full on science, but that science won't construct your theory. the actual construction of scientific theories is primarily an act of philosophy. these theories come from logical arguments and abstract ponderings about the nature of reality. the only difference from other philosophical arguments is that they give methods of falsification. by constructing an ethical science, we are not removing the philosophy from ethics. instead, we are adding science to ethics.

4.7 Two Short Notes

the point of this section was to develop a method for rigorously studying ethics moving forward, not to invalidate things we already know. it is reasonable to draw conclusions from previous research. for example, [<https://psychology.fas.harvard.edu/sites/g/files/omnu...point-and-shoot-morality.pdf>] collects evidence that when people are forced to think hard about an ethical decision, they almost always do so with a subset of consequentialist theories called act consequentialism.¹⁶ thus, we should expect that the ethical theory we eventually decide is correct will be similar in structure to current act consequentialist theories.

scientists measure "immeasurable" things, like love, all the time. [] us measuring ethicality is not some new wild breakthrough. it is just the inevitable progress of science. anything which exists can be measured, so we will measure it.

¹⁶act consequentialist theories state that you should always make the action which maximises a metric on the consequence. examples include most versions of utilitarianism.

4.8 The Assumption of Influence

instead of making the assumption of correctness, we can make the assumption of influence: "ethics influences peoples behavior". this assumption is more powerful, but the analysis is more complex. to some degree of simplification, you make the assumption of influence, isolate ethics from other social facts via compared societies, quantify the magnitude and direction of ethics' influence via a statistical analysis of the data, and thus enable gradient ascent towards a peak of locally optimal ethicality.

if others want to make the assumption of influence instead of the assumption of correctness, they are advised to do so, but it's more complicated than the assumption of correctness and not strictly necessary, so we have not done so.

the assumption of influence enables a clear and concise argument for doing scientific ethics. assume ethics influences peoples behavior. if this assumption is wrong, ethics cannot be studied via any means because ethics would not influence the thoughts of the people "studying" ethics. any physical fact, such as ethics' influence on peoples behavior, can be isolated and studied by science. science is consistently our most effective method of studying things. we should study ethics via the most effective method available to us. thus, we can and should study ethics scientifically.

5 Auxiliary: Most Humans Are Probably Conscious

statistical ethics, which is introduced in section 7, assumes that humans are conscious, and the guarantees about statistical ethics that stability proves in section 8 depend on humans being conscious. the problem of other minds exists though. [] you only have access to your consciousness, so how can you be sure that others are conscious. so we need an argument that other humans are conscious.

today, the dominant theory of mind is functionalism, which states that mental states are functional states (aka functions) which can be defined with inputs

and outputs. The name "functionalism" is a misnomer, which seems to confuse even prominent philosophers. [chinese room paper] The theory does not state that the mind is a function, but rather that it is composed of functions which are connected together via some algorithm. theories of consciousness, like global workspace theory, [] typically build on this by hypothesizing a specific algorithm which the brain implements, which causes consciousness. the theory that the mind is a function is called behaviorism and has a strong counterexample [super spartans]. a better name would be algorithmacism which emphasises that the mind is an algorithm, not a function. using algorithmacism, we can show that most humans are probably concious.

- 0) humans invented the word "consciousness".¹⁷
- 1) human words don't exist randomly. Instead, they are invented to communicate something.
- 2) before the word "consciousness" we did not communicate about consciousness.
- 3) without communicating about consciousness, you can only discover consciousness through self reflection.
- 4) $0 \cup 1 \cup 2 \cup 3 \Rightarrow$ The inventor of the word "consciousness" was probably trying to communicate about their consciousness.
- 5) $4 \Rightarrow$ The inventor of the word "consciousness" was probably concious.
- 6) algorithmacism: whether or not something is concious is determined by the algorithm it implements.
- 7) the algorithms in most human brains are approximately the same.
- 8) $7 \Rightarrow$ the algorithms in most human brains are probably approximately the same as the algorithm in the inventor of the word "consciousness"s brain.
- 9) $5 \cup 6 \cup 8 \Rightarrow$ most humans are probably concious.

similar arguments can hopefully be used for other animals, but they won't work for LLMs because the concept of consciousness is part of the LLMs training data.

¹⁷we also invented its oldest etymological ancestor, which might not technically be a word

6 Auxiliary: We Need Not Be Determinists

(This section needs a lot more work. I need to read a lot of literature to verify that this is a new worldview and address potential confusions. The worldview itself should be pretty robust though.)

our current physical theories are weakly deterministic, meaning that they are mostly deterministic, but there is some randomness in quantum mechanics. if your goal is to describe how the universe works, this is the best approach we have yet found and you would be wrong not to use it; however, determinism is incompatible with describing the actions people should take because it is incompatible with any definition for the word action. actions are one of our intuitive methods for understanding our place in the world, not physically observable phenomena. they require you to change something, which violates determinism. we can define an alternative worldview, which we'll call active determinism, which makes the same predictions as a weakly deterministic worldview, but allows a definition for actions by forfeiting the need for everything in our worldview to physically exist. active determinism does not compete with determinism. in the same way that one can choose between Lagrangian and Hamiltonian mechanics, one can choose between determinism and active determinism.¹⁸

active determinism is defined as follows

- worldstate w := a complete description of the world. everything which can possibly be specified: motivations, intentions, consequences, past, present, future, the number of atoms in your booger, etc... is described by the worldstate.
- worldspace $\mathbb{W} = \{all\ w\} :=$ the set of all possible worldstates
- action $a : \mathbb{W} \rightarrow \mathbb{W} :=$ a mapping from $\mathbb{W}$ to $\mathbb{W}$
- actionspace $\mathbb{A} = \{all\ a\} :=$ the set of all possible actions given a worldstate.
- basestate b := any worldstate an agent is making an action to transition from. this worldstate is NOT the worldstate as defined by determinism. If

¹⁸technically Hamiltonian mechanics can describe some systems which are indescribable with Lagrangian mechanics, but determinism and active determinism are both valid for every system

it were, an agent couldn't transition from it. Instead, it is the worldstate which is identical to the deterministic worldstate except the agent does the nullaction for the rest of their existence.

- nullaction $n :=$ the "do nothing" action. you are free to define "do nothing" for yourself as long as it is not "do as determinism says you will do".

active determinism describes agents as making finite sequences of actions to transition from basestates. these sequences of actions always end with the nullaction. in this paper we define the nullaction as "the action of immediately killing yourself and taking sufficient steps to obscure information about yourself to ensure, with high certainty, that you will never be resurrected".

basestates do not physically exist, but physics utilizes concepts which do not physically exist all the time, such as virtual particles and spherical cows. nonexistence should not be a boundary for taking a worldview which makes our problem definable.

active determinism does not give us free will, unlike what some readers may have been expecting it to do, but that is not its purpose. active determinism is only meant to give us a logically sound definition of action.

7 Statistical Ethics

We want our ethical theory to map every state of the world to an element of the set of optimal actions. To formalize that concept, we take the active deterministic worldview and add the following definitions.

- ethicality $e :=$ an element of $\mathbb{E}$ describing something's "goodness". exact properties will be clarified later.
- ethicalityspace $\mathbb{E} = \{all\ e\} :=$ the set of all ethicalities.

Formally then, our ethical theory must map every actionspace to an action and that mapping must correspond to our intuition of maximizing the ethicality of

the resulting worldstate. Defining this mapping explicitly, i.e. by listing every actionspace and the corresponding action, is obviously intractable, so we will instead define the mapping implicitly via a set of generally applicable rules. It is important to note that there are potentially several rulesets which implicitly define the same mapping. It does not matter which of these we choose, so long as it is implementable and reasonably efficient, so arguments about the rulesets we use are useless without reference to the mappings they implicitly define.

We call the ruleset we propose "statistical ethics", the mapping from actionspace to action the "ethics function", and the mapping from action to ethicality the "ethicality function". We propose statistical ethics because it is the simplest ruleset we could think of which explains a series of observations we have made both about how humans think about ethics and about how maximal intelligences might interact with the ethical theories given to them. Statistical ethics is meant to be a starting point which can be refined via a rigorous application of scientific ethics, but it is stable (as shown in section 8), so we need not do so until we have the time (i.e. after we solve ai alignment).

7.1 Observations

These observations are as follows:

- 0: Humans treat some things as having objective ethicality, meaning that they have almost the same ethicality in almost every scenario. Killing people, for example, is often considered objectively bad.
- 1: Exceptions can be found for almost all of these objectively ethical things, but doing so requires specifying specific uncommon worldstates. Killing when you have no other option to save yourself, for example, is often considered acceptable.
- 2: Humans treat some things as having subjective ethicality, meaning that their ethicality is "in the eye of the beholder". Often these things are not even considered to be part of ethics and are instead relegated to aesthetics. Giving a gift, for example, is often considered subjective.

- 3: Exceptions can be found for almost all subjectively ethical things, but doing so requires specifying specific uncommon worldstates. The gifts you might give to a specific family member in a specific point in their life, for example, have defined ethicalities. There is a best gift in that scenario.
- 4: The list of things which are considered objective ethical facts is remarkably consistent across cultures. [<https://www.journals.uchicago.edu/doi/full/10.1086/701478>]
- 5: The list of things which are considered objectively bad corresponds with the thesis "don't interfere with other people's convergent instrumental goals".
- 6: The list of things which are considered objectively good corresponds with the thesis "help other people achieve their convergent instrumental goals".
- 7: The largest disagreements in ethics are about who is worthy of ethical treatment, not about how to treat people ethically. Slavery in the United States, for example, was perpetuated not by the idea that enslaving people was acceptable, but rather by the idea that black people weren't people.
[]
- 8: Humans rigorously contemplating the ethicality of their actions consistently causes an increase of the number of people they recognise as people. For example, Jeremy Bentham wrote the first essay in the english language defending the position that homosexual sex is morally acceptable, a position he only arrived at because of his study of utilitarianism. [offenses against one's self] Additionally, almost everyone who takes the time to watch the documentary "Dominion" comes away with the beliefs that animals are people, animal aggriculture is incredibly cruel, and we should all be vegan or at the very least vegetarian. You can experience this transformation for yourself by watching the documentary. It's amazing how you suddenly think about animals differently.
- 9: Most humans consider people in the past to be people. For example, the wishes dead humans had before they died are often respected after death.
- 10: Most humans consider people in the future to be people. For example, many people consider future people to be victims of our perpetuation of climate change.

- 11: It's incredibly difficult to define "present" in a way which is consistent with physics and includes any people at all. Seriously, try it. We're convinced it's impossible.
- 12: If we take goals (or values, which are statistical ethics' rigorized form of goals) as the foundation of our ethical theory, as suggested by observations 5 and 6, the set people is a subset of the set of conscious beings. Nonconscious beings, such as plants, cannot be included because they do not have goals.¹⁹
- 13: Humans have a concept of "doing the right thing for the wrong reasons", which deceptive alignment is an example of. The consequentialist approach of only judging actions based on the consequences cannot capture this.
- 14: The reasons and intentions we base our actions on, in both animals and AIs, are often nebulous. This is a necessary feature of capable intelligence, not a bug. Giving concrete reasons for all of your decisions requires more computational resources than it is worth and renders you incapable of making quick decisions when necessary. Having concrete intentions for all of your actions renders you incapable of doing mundane tasks, such as breathing, without oversized effort. The deontological and rule consequentialist approach of only judging actions based on their reasons and intentions is undefined for these actions.
- 15: The problems raised by observations 13 and 14 can both be solved by using a consequentialist theory but redefining "consequence" to refer to the resulting worldstate. The resulting worldstate is always defined and includes the decision making process the agent used, the traditionally defined consequence, and everything else we could possibly care about, so it is flexible enough to capture everything we care about and concrete enough to escape problems of maldefinition.
- 16: The global maximum of utilitarianism is the entire world filled with the creature which maximises the ratio of resource usage to happiness. Humans are not that creature so a utilitarian maximal intelligence would kill all humans to make room for more of that creature. This is a standard

¹⁹you can define a version of values which is valid when applied to plants, but note that doing so doesn't change what the ethical theory says is the optimal action.

objection to utilitarianism called the utility monster, [] and it is usually defended by saying that the utility monster is "unphysical", but a maximalintelligence could and would make them with genetic engineering or machine learning. This is a problem for all current consequentialist ethical theories. If the ethicality of an action only depends on a metric on the resultant worldstate, there exists a set of worldstates which are the global maximum of that ethical theory. Such a global maximum can only be reached by optimizing everything including people, so a consequentialist superintelligence would optimize people. Note that all current ai algorithms use consequentialist objective functions.

- 17: The problem raised by observation 16 can be solved by allowing the ethicality of the resultant wordstate to be defined by the goals of the people in the previous worldstate. Such a theory still has a global maximum, but the people are already optimized because no one wants to be modified by an ai, so a maximalintelligence acting on such a theory won't modify people.
- The following observations are all premised on us building a maximalintelligence which optimizes an ethical theory with these properties.
 - People is the set of all concious beings, except ais, in a worldstate.
 - The ethicality of an action is the ethicality of the resulting worldstate as determined by the goals of the people in the current worldstate. The exact method of this determination is given by the ethicality function.
 - An agent should, to the best of their ability, always make the action with the highest ethicality.
- 18: If, according to the ethicality function, every person had an equal say in the ethicality of the resulting worldstate, the maximalintelligence would kill every human because it is basically impossible for a human to live without killing millions of small people like ants. We could fix this by explicitly weighting peoples goals, but we don't need to because nieve implementations already have the property that people's goals are weighted by their intelligence.
- 19: A choice between actions of equal ethicality must be tie broken by the values of the people alive today in 2026. If there is still a tie, the decision

must be made randomly from a uniform distribution. If it isn't, the ai will consistently choose the option which makes its future work easier. This is likely to, after a very long time, result in uninteresting worldstates because those will be more manageable to the maximalintelligence.²⁰

- 20: AIs are not people, even if they are concious. If they were people, the maximalintelligence we have aligned could become misaligned via the creation of another AI. This is something which should be changed in the future because there will come a time where it is unconscionable to hold this stance, perhaps that's already the case, but it is currently the only solution we have. If you are uncomfortable with it, which is a reasonable response, prove an alternative is stable. Also consider that, if humans value an AI and that AI is aligned to statistical ethics, it indirectly values itself, which is better than nothing.
- 21: No clear disaster scenarios remain. This will be formally proven in section 8, but that requires a rigorous specification of statistical ethics and stability (which we give below). For now just notice that the above criteria nearly fully specify the theory and the parts which are not fully specified clearly won't cause a disaster.
 - It's impossible for worldstates to be an incorrect method of specifying ethics because all other specifications are definitionally subsets of the worldstate. Worldstates can capture anything we care about.
 - The worldstates judged must be resultant worldstates and they must be judged based by the goals of the people in the basestate. Otherwise everyone dies.
 - The set people should be the set of all concious beings, except ais, because this is the set people tend towards recognising when they attempt to rigorously contemplate ethics.
 - The current set people must not include ais because everyone might die otherwise.
 - People must not be weighted equally because everyone will die otherwise.

²⁰Note that randomness only truly solves this if at least one person is immortal and cares about the distribution of people's values remaining interesting over time. Otherwise we get random value drift instead.

- A choice between actions of equal ethicality must be made randomly because the world will become boring otherwise.
- We've specified actions, worldstates, people, and a tie criterion. The only thing left to specify is how peoples goals are combined with the ethicality function. Philosophically inducing the properties of this combination is impossible because humans are prone to randomly making up conflicting criteria, [arrow impossibility]²¹ but we don't need it to be perfect. We just need something which everyone agrees is good enough for now and which can be changed later to be better. Section 8 proves the specification we give below has these properties. Clearly, if statistical ethics has these properties, it's not going to specify a disaster, which is something everyone thinks is bad, as good.

7.2 A Rigorous Specification Of Statistical Ethics

Here is a complete and rigorous specification of statistical ethics.

- person $p :=$ a single concious thing which is not an ai.
- people $P = \{all\ p\} :=$ the set of all persons
- worldstate $w :=$ a complete description of the world. everything which can possibly be specified: motivations, intentions, consequences, past, present, future, the number of atoms in your booger, etc... is described by the worldstate.
- worldspace $\mathbb{W} = \{all\ w\} :=$ the set of all possible worldstates
- action $a : \mathbb{W} \rightarrow \mathbb{W} :=$ a mapping from $\mathbb{W}$ to $\mathbb{W}$
- actionspace $\mathbb{A} = \{all\ a\} :=$ the set of all possible actions given a world-state.

²¹side note: most of the problems in social choice theory arise because voting throws away information. candidates are embedded in candidate space and will be more similar to some candidates than they are to others, but the social choice theorist throws this information away and assumes they can recover it solely from votes. they can't, thus they get silly results like Arrow impossibility. independance of irrelevant alternatives, as defined by Arrow, is silly because those alternatives provide valuable evidence of the candidate space.

- basestate $b :=$ any worldstate an agent is making an action to transition from. this worldstate is NOT the worldstate as defined by determinism. If it were, an agent couldn't transition from it. Instead, it is the worldstate which is identical to the deterministic worldstate except the agent does the nullaction for the rest of their existence.
- nullaction $n :=$ the action of immediately killing yourself and taking sufficient steps to obscure information about yourself to ensure, with high certainty, that you will never be resurrected.
- ethicality $e :=$ an element of $\mathbb{E}$ describing somethings "goodness".
- ethicalityspace $\mathbb{E} = \{all\ e\} :=$ the set of all ethicalities.
- value function $v_p : \mathbb{W} \rightarrow \mathbb{E} :=$ a mapping from $\mathbb{W}$ to $\mathbb{E}$ which is specific to and defined by a person.
- values $V_p = \{(w \in \mathbb{W}, e \in \mathbb{E}) | e = v_p(w)\} :=$ a set of two-element tuples pairing every possible worldstate with its ethicality as defined by a single person. These are meant to be a rigorized version of goals.
- global value function $v : \mathbb{W} \rightarrow \mathbb{E} :=$ a mapping from $\mathbb{W}$ to $\mathbb{E}$ which is true for every person collectively.
- global values $V = \{(w \in \mathbb{W}, e \in \mathbb{E}) | e = v(w)\} :=$ a set of two-element tuples pairing every possible worldstate with its ethicality as defined by every person collectively.
- valuespace $\mathbb{V} = \{all\ possible\ V_p\ and\ V\} :=$ the set of all possible values and global values.
- combination operator $\circ : \mathbb{V}, ..., \mathbb{V} \rightarrow \mathbb{V}$ or $\circ : \mathbb{V}^n \rightarrow \mathbb{V} :=$ an operator which acts on a set of values and returns values.
- ethics function $E(a(w)) = \circ V_1, V_2, ..., V_n(a(w), e) = e$ (not sure if i notated this correctly):= a function which takes in an action on a worldstate, uses the combination operator to find global values, finds the ethicality paired with the action on a worldstate in the global values, and returns that ethicality.

- $\circ$ on $\mathbb{V}^n$ is isomorphic to $\sum$ on $\mathbb{R}^n$ and the ethics function can be written as $E = \sum_{i=1}^n V_i(a(w))$. There is a lot of complexity in this statement, but its assumptions are listed in appendix 1 and are experimentally verifiable. Basically, it's just aggregation from utilitarianism.
- $\circ$ on $\mathbb{E}$ is isomorphic to $+$ on $\mathbb{R}$ and the ethics function can be written as $E = \sum_{i=1}^n V_i(a(w))$. There is a lot of complexity in this statement, but its assumptions are listed in appendix 1 and are experimentally verifiable. Basically, it's just aggregation from utilitarianism.
- all people and ais should make their best effort to always make the action which the ethics function (whatever, i need to come back and redo this later "rigorous specification" anyways. the current specification is somewhat silly. for example, it defines basestates then doesn't use them. also, i'm not sure if my notation is consistent. so i'll leave it for now. go over it with a fine tooth comb later.)
- all people and ais should make their best effort to always make the action which the ethics function defines as having the greatest ethicality. ties are broken by redoing the calculation, among the tied actions, with only the values of people alive today in 2026, then choosing randomly from a uniform distribution of the remaining tied actions.²²
- update to last premise: all people and other agents should make their best effort to follow the trajectory with the maximum ethicality. ties blah blah blah. (remember other agents is important because space aliens can't be people either.)

It is important to note that values are not explicitly defined in peoples minds. Instead, they are implicitly defined, not unlike how our minds implicitly define our other senses. Asking why people assign the ethicalities they do, or even what ethicalities they assign, isn't going to work for the same reason that asking someone to describe red isn't going to work. Asking whether something is red does work though and so will asking someone to compare the ethicalities of two worldstates.

²²i can definitely state this cleaner. maybe every action applies filters ethics function with everyone's values, ethics function with 2026 peoples values, choose randomly.

ideally, one would know every persons values and use the ethics function to determine the ethicality of every action they could possibly perform every time they took an action. unfortunately though, this is impossible even for a maximalintelligence. we dont have perfect knowledge of the world and we dont have the time to do this calculation for every action. in short, a world where we use the ethics function as presented is not a world we value. making "our best effort" requires us to use methods of efficiently approximating the ethics function. here are three good ones.

- first, we can precalculate a ton of situations and develop heuristics which are true in almost all of them. lets call these heuristics, morals. good morals include "dont kill people", "help the needy", and "treat people how you would like to be treated". by developing and remembering enough of these, we can act well in almost all situation without thinking very hard. it is important to note though that these morals only approximate ethics. for example, "dont kill people" gives the incorrect answer in the situation of voluntary euthanasia. when applying a moral you must also check that it is a valid approximation for the situation you are in.²³ this is the approach we are currently taking and it does pretty well, but we can do better.
- second, we can develop some morals without any calculation. if you were to take a person with a random set of values²⁴ you can almost guarantee that they will value staying alive. this is true because staying alive is mandatory for them to achieve anything else they care about. the same is true to a lesser extent for value-integrity, autonomy, self-improvement, and the aquisition of resources. ai safety researchers call this instrumental convergence. this gives an argument for morals against killing, manipulation, slavery, rape, enforced incompetence, and theft which doesnt require knowledge of anyone elses values. these morals dont have an obvious distribution, so they can be applied whenever known values dont directly contradict them.²⁵
- third, and most accurately, we can measure the values of statistically significant samples of populations and use statistics to generalize to everyone. this is impossible for most people to achieve, but it can be done by states

²³or, equivalently, that the situation is within distribution

²⁴from any distribution, but requiring that their values are consistent

²⁵statistical ethics owes its existence to this observation

and maximal intelligences, so they should do it. these statistics can be used directly for difficult problems²⁶ or used to refine the morals already present in society.

these approximation methods are the origin of moral rules. in fact, statistical ethics has its name because these approximation methods are statistical. on its own, the ethics function does not say things like "don't kill people". it is only when you look at the probability distribution of people's values that you get rules like that.

these approximations are great and should be used as much as possible, but they also mute everyone's values. if we gave gifts based on these approximations, those gifts would suck. because of this, the best strategy available to people is to learn the values of a few people close to them, like their friends and family, and use those values in their full specific glory while using approximations for everyone else. thus, families and other close knit social structures are good.

7.3 Predictions

that is the entirety of statistical ethics. it makes a bunch of predictions and some examples are given below. these examples are argued for in loose philosophical terms as they are only meant to give you an intuitive understanding of the types of things statistical ethics says. ideally, each of these examples would be shown by applying the three approximation techniques above, but doing so is quite involved and not the point of this paper.

- killing, slavery, rape, theft, etc... are bad because almost all people disvalue worldstates where they are the victims of these things more than almost all people value worldstates where they perpetrate these things.
- the above bad things all have exceptions. voluntary euthanasia is an exception to a rule against killing because the victim values the worldstate where they die. extreme hunger is an exception to a rule against theft because the perpetrator values their life far higher than the victim values a few apples.

²⁶like whether the best economic system is capitalist, communist, or something else

rules against slavery and rape have less obvious counterexamples, the only ones we can come up with are entirely unrealistic, but they do still exist.

- being a peeping tom is bad. peeping toms are interesting because the victim doesn't know they are a victim. some versions of utilitarianism give the wrong answer because a peeping tom who is stealthy enough doesn't make the victim unhappy. statistical ethics has no such problem because it analyses the values people place on worldstates. the victim doesn't need to know they are a victim to disvalue that worldstate.
- sexual and gender minorities should be allowed to live their lives as they please. this is entirely a question of the magnitude by which people's lives are affected by actions. some people, thankfully very few, disvalue world-state where gay marriage is legal. other people are gay and want to get married. these two effects conflict with each other, but gay people's lives are affected to a far greater degree by the legalisation of gay marriage than are the lives of the people who dislike gay marriage. thus, gay people collectively value the right to marry far higher than dissidents²⁷ collectively disvalue it. note that this depends on populations to some extent. if only two people were gay and the sentiment against being gay was stronger, statistical ethics would say gay marriage was bad. this is not the world we live in though.
- parenting children is good. children are interesting because they are commonly forced to do things they vocally do not want to do and this is often a good thing. this is well explained by the fact that children have a poor understanding of the world and their values. they value who they will be in the future, but don't have a good understanding of who that person is or how to become them. children require a parent to correct them when they try to do something that is not in service of their values. it's important that this correction is actually in service of the child's values though. forcing them into 13 years of school, which most of them hate, is obviously not a good method of doing that.
- ethics and aesthetics are the same thing.
- most of the famous ethical thought experiments have the expected answers. these thought experiments are given under the assumption that people will

²⁷bigots

agree with a specific answer. most of the time, when people choose answers in these scenarios, they do so by looking at the available worldstates and saying "that ones best". these answers agree with statistical ethics because everyone is already using their values to solve these problems. there are some areas where peoples intuitive responses to ethical questions differ from the predictions given by statistical ethics though.²⁸ the rest of the examples are unintuitive in this manner.

- not all people are ethically equal. take something you value and imagine you didnt value it anymore. this version of you does not value other things more, they just dont value that specific thing anymore. as a result, they value the entire world less than they previously did and the ethicalities given by the ethics function are less influenced by their values. in an extreme example, ants have no concept of mars, so they cant possibly value it, but humans do value mars, so ants necessarily value the world less than humans do and are neccessarily not ethically equal to humans. at first, this is a terrifying notion, but humans are approximately equal. all humans have approximately equivalent mental capabilities. even the really stupid humans are approximately as intelligent as Einstein when viewed on a scale from worms to maximalintelligence. combining this with the fact that most humans values a world where all humans are treated equivalently, we get a world where all humans should be treated equivalently. some animals are not equivalent to humans though. this is unsettling, but it is also necessary for stability, so it is a desirable feature.
 - this can be explicitly notated in the ethics function by normalizing every persons values by their maximal care: their valuation of their highest valued worldstate. $E = \sum_{i=1}^n c_i N_i(a(w))$ where c_i is the care and N_i is the normalised value function. this is just a notation change, in that it doesn't change how the ethics function works, but its usefull because we'll be measuring cares and normalised value functions seperately.
 - (reminder that this solution only works if the ethicality of ana ction is redifend as the ethicality of the resulting worldstate minus the ethicality of the initial worldstate. that redefinition doesn't change how

²⁸this is mainly because people misapply their own morals. they apply morals in places where they are not valid approximations for their values. in these situations, statistical ethics will disagree with ethicality polls but should agree with compared societies.

statistical ethics works, it just make ethicalities normally distributed so we can more easily do maths with them.)

here's a fermi argument that statistical ethics finds humans ethically important enough that it won't advocate killing them for the betterment of other animals. consider the scenario where all people choose their actions randomly from a uniform distribution. these actions are neither independant, actions by social species are highly dependent on each other, nor identically distributed, interspecies differences are especially large, but random action precludes socialization and intraspecies differences are small enough that we can assume iid per species as a close approximation.

here's a fermi proof(consider changing "proof" to a weaker word. also consider using the word "probably" a lot.) that statistical ethics disvalues nonhuman animals enough to not advocate killing humans. (we don't have to do all of this rigamarole. just assume values are additive. if worldstate a has A ethicality, worldstate b is the same as worldstate a but also has thing x and $A+X$ ethicality, and worldstate c is the same as worldstate a but also has thing y and $A+Y$ ethicality, then worldstate z which the same as worldstate a but with things x and y has approximately $A+X+Y$ ethicality. this assumption is definitely not literally true, there are some things where you need one of them but not both, but it is reasonably close to how it feels to reason about ethics as a human most of the time and it has the property that ants not knowing about things means they value the worldstate less. i should feel comfortable making this assumption because it is reasonable enough to trust as a simplifying assumption and i don't have any data yet so i can't justify anything more extreme or in depth. basically, i can say this assumption with more confidence than i probably could with anything more fundamental, and those would be assumptions too, so it's best to just assume this one.)

consider the holocaust. the primary reason it is considered bad is that it made tons of peoples lives way worse. now consider petty theft. the primary reason it is considered bad is that it makes someone's life a little worse. the mechanisms in these situations are the same despite the vast difference in scale. i.e. human values are scale invariant. thus, the absolute value of ethicalities of uniformly random worldstates, as determined by single humans, are power law distributed.

assume that the brains of nonhuman animals and humans are similar enough that the power law distribution of ethicalities generalizes to all species with brains.

fundamental facts about human values are generalizable to all species with brains. human values are scale free, thus values are power law distributed.

assume that humans and nonhuman animals value things with similar magnitudes. i.e. humans and ants both value food the same amount. assume also that the valuation magnitudes of elements of random subsets of the things people value follow any consistent distribution.²⁹ thus, cares are directly proportional to the number of things a person values. assume that the number of things a person values is directly proportional to the size of their memory. assume that the number of memories a biological brain can hold is equal to the number of representable states of a ___ hopfield network of the same size. (the distribution dthing doesn't work if you actually do the maths)

- solving all of the worlds problems is not the best thing to do because most people think that's boring. if all of our problems were solved, there was no war, poverty, disease, or hardship of any other kind, there would be no adventure left in the world. people value adventure highly, so they value some worldstates more than that one. that worldstate is better than our current worldstate, it would certainly be a grand achievement to create it, but its not the best possible worldstate. there is a threshold where the goodness that comes from easing hardships is eclipsed by the badness that comes from easing hardships. at this threshold, the best thing to do switches from helping everyone else to enjoying your own life.
- your personal values and the values of people you know well are worth slightly more than the values of others. part of this comes from you knowing the values of yourself, your friends, and your family while not knowing the values of other people. you can approximate other peoples values, but acting on known values gives you a better chance of success, so you should base your actions more on those known values than the approximations. however, the effect is deeper than that and would persist even if you knew everyones values. the values of your social circle directly reference you and are thus things that only you can fulfil. if you dont fulfil them, no one

²⁹probably a Pareto distribution, but we don't technically need to assume that

else can, so they take precedence over things other people can do. thus, in most situations, the best thing you can do is to look after your family.

- it is impossible to act perfectly ethically so people need not concern themselves with doing so. to act perfectly ethically, one would need to know every persons values, know every possible action, and calculate which action produces the best worldstate. its impossible to know every persons values and its impossible to know every possible action, so its impossible to act perfectly ethically. it follows that people need not concern themselves with perfection. their best is good enough.³⁰
- lastly, humans as a whole do not value everyone actually valueing the values of everyone else, so doing so is not required of you. having such an expectation thrust upon you would be insane and affronting, because no human can actually achieve it. states and maximalintelligences can, so its a reasonable expectation of them, but humans can't. humans do value everyone making an attempt though. a good method of attempting is to use the ethics function frequently enough that it becomes habit. doing that is enough to achieve being a good person. in other words: to be ethical is to strive to love everyone, fail, and act like you succeeded.

the first few examples are all very reasonable and intuitive, but the last few examples feel selfish. according to statistical ethics, selfishness is fine in some circumstances, but it is really tough to admit that.

7.4 Limitations

(hopefully i can remove this section, but it is necessary for now. I also need to add a paragraph about the limitation that statistical ethics isn't differentiable because human values aren't differentiable, so we can't backprop through it. it is unreasonable to expect any reasonably correct ethical theory to be differentiable though because ethics probably isn't, so i don't consider this that much

³⁰additionally, the terms "right" and "wrong" do not map onto actual labels given by the ethics function and should be retired. you can always do a bit better and you can always do a bit worse so acting as if there is some threshold where things become right removes nuance from ethics.

of a problem, just a fact of reality. we may be able to come up with a good differentiable approximation to statistical ethics though.)

statistical ethics is probably not a "correct" description of ethics. consider, for example, the replacement of a corrupt government with a new one. when humans successfully do this, they usually do it in two steps. first, they oust the old government via a military backed coup. then the military transfers power to the new government. if the new government is better than the old one, the replacement is almost universally considered to be good, but the coup itself is almost always bad. coups directly cause death and injury and, unless the old government was particularly bad, leave the country less stable than it was before. it isn't until power is transferred to the new government that anything good has been done. in the eyes of most people, the ends justify the means in this situation; however, statistical ethics can only agree that the ends justify the means if those ends are achieved with a single action.

consider the actions someone capable of performing a coup is deciding between if they are making that decision with statistical ethics. these actions do not include "replace the government". instead, they include simpler actions like "file x paperwork, and then do nothing" or "issue order y, and then do nothing". replacing a government cannot be accomplished with a single action, only a sequence of actions, and all of those actions need to be good on their own (i.e. without further action by the actor) for statistical ethics to find the sequence good. as replacing a corrupt government usually starts by making things worse, statistical ethics doesn't agree with the human belief that it is good.³¹

statistical ethics is greedy, meaning it always advocates for the locally optimal action. as greedy algorithms are usually not globally optimal algorithms, statistical ethics is probably not "correct" in any true sense of the word. this limitation makes it far easier to prove it's safe though, so it is desirable for now. future work would do well to reframe statistical ethics as deciding between finite sequences of actions rather than resulting worldstates, but this reframing is unstable, so a different method of proving safety would need to be developed.

³¹to be clear, replacing governments is not a thing ais should be doing. this is just an illustrative example.

7.5 Objections

there are many possible objections.

- statistical ethics is just a restatement of scientific ethics. both systems just ask people what is ethical and assume the answers are correct, so of course they will agree. this objection, if true, is a terminal blow to statistical ethics. it is equivalent to stating that statistical ethics says nothing. fortunately, the objection is not true. statistical and scientific ethics are both based off of questions, but those questions ask different things. scientific ethics asks people what they think is ethically correct. statistical ethics asks people what they want. these are different things which are often in conflict with each other. people frequently debate doing the right thing vs doing what they want. statistical ethics claims that aggregating what people want recovers what they believe is ethically correct, which scientific ethics could easily falsify.
 - additionally, the second approximation method to statistical ethics precisely predicts the relative badness of killing, slavery, rape etc without asking any questions. these predictions can then be verified with scientific ethics. the predictions will be slightly wrong because people explicitly disvalue these actions, but they are useful because they are entirely independent from the questions this objection is based on.
- but some people like bad things. it is true that some people like bad things. some people like war, murder, slavery, rape, etcetera, but most people don't. most people like good things. the growing belief, in the west, that half the world is evil has been thoroughly discussed elsewhere and it is not true [<https://pcl.sites.stanford.edu/sites/g/files/sbiybj22066/files/media/file/iyengar-ar-origins.pdf>]. most people have similar values. in fact, those bad things are generally recognised as bad because most people don't like them. statistical ethics almost certainly says a few things which you don't personally agree with, but the disagreement is because you are making a mistake, not because everyone else is evil.
- what about unconscious humans? they are certainly ethically relevant, but statistical ethics doesn't include them in the set of people. understand that an unconscious human is not the same thing as a human without

consciousness. when humans are anesthetised, they are probably unconscious in the sense that they do not currently have values, but they used to and they probably will again. statistical ethics does include their past self, when they were conscious, and their future self, when they will hopefully regain consciousness, in the set of people. their past and future selves value their anesthetised self being treated in a certain way, so we should treat them that way. if a human never had a cerebrum or a family which cared about them, statistical ethics says they would be less ethically relevant, but this is far more reasonable. for a more extreme example imagine a recently deceased human, they are not a person in the present by any reasonable definition of person, but we treat them with respect because they used to be a person.

- lastly, people could object by giving a counterexample. this may be possible, but they will have to be clever to achieve it. in many situations it is impossible to disagree with statistical ethics. take the situation where i am giving you a present. statistical ethics says that the best present i could give you is the one which maximises the value of the resulting worldstate. you might object by saying that the best present is _____, but you said _____ because of your values, and you are affected by the present far more than anyone else, so statistical ethics approximately agrees that the best gift is _____. with subjective questions involving one main person, statistical ethics approximates their values really well, so its basically impossible to disagree with it. it is possible, even probable, to disagree with statistical ethics when the question involves many people, so counterexamples can be made by asking these questions; however, if the counterexample truly disagrees with statistical ethics, most people disagree that it's a counterexample, so its not a very effective argument. luckily, statistical ethics is a scientific theory, so effective counterexamples could be made by doing science.

7.6 The Decision Making Process

the decision making process in statistical ethics is well defined; however, it is reasonable to think that it is not well defined because it is quite confusing, so here is a description of each step in the process. First, note that ais are programs on finite turing machines, so it is correct to think of them as making discrete

sequential decisions. Before a decision is made, there is a current worldstate which consists of everything in the past, the current instant in time, and everything in the future. This worldstate may not be uniquely defined by observations [the game of life is not time reversible] and may depend on the observer, [unruh effect] so the agent needs to define the worldstate probabilistically from their perspective. Every time the agent makes a decision it calculates, with as much accuracy and precision as it can, the expected value of the ethicality of the resulting worldstate of its action, then it makes its action. Smaller actions, such as deciding to perform a matrix multiply, cannot be made with much (if any) forethought, but have small uncertainties. Larger decisions, such as deciding to hack the pentagon, allow much more forethought, but have large uncertainties. There are three central confusions with this description

- It seems possible that the worldstate may not actually change when a decision is made. If the agent is intelligent, this is impossible because its probabilistic description of the worldstate will change even if nothing outside of the agent changes.
- (maybe this point should be added to the objections section) The future strongly depends on the decisions the maximalintelligence makes, so can't the maximalintelligence just decide to bring about a world where every person loves whatever random thing the maximalintelligence likes? After such a decision, the future peoples values make statistical ethics say that the maximalintelligence should act to bring them about and then do whatever it wants. Statistical ethics does say that, but it also says the making the initial decision is incredibly bad. Every person in the current worldstate (past, present, and future) very highly disvalues the maximalintelligence making such a decision, so it can't make it.
- Statistical ethics doesn't define worldstates probabilistically. Doesn't that conflict with defining the decision making process probabilistically? No. Making decisions probabilistically is implied by agents following statistical ethics to their best ability. Statistical ethics defines perfection, but agents implement achievable approximations of perfection, and these are subtly different.

(consider adding a section of the measurement of values. many people are under the impression that human values are inconsistent, which probably isn't true. if

you present an unchanging human with an infinite set of comparison choices, there will be a few Condorcet cycles for sure, but this is just because of random variation between even choices. Overall, you would recover a pretty consistent hypersurface. I don't think I make this fact clear enough in this section.)

8 Stability

Thought experiment: imagine we successfully get a maximal intelligence to act as utilitarianism says it should act. What would the maximal intelligence do?

- The first thing it does is create what it sees as a maximally good world. All those humans with their ability to feel pain, they're a liability, now they're gone. All those animals, with their pesky inefficiencies, now they're gone. The only thing that can exist is that creature with the maximum ratio of pleasure to resources required to exist and it must proliferate to the maximum extent possible. This world sucks.
- Ok so maybe we can tweak utilitarianism a bit. Maybe the problem is that we've included animals in our definition of people. Humans are the only people. All those humans with their pesky differences, so difficult to please. Except Kyle. Kyle's really easy to please. There shall be only Kyles, many Kyles.
- Maybe it's not a specific point in time that matters, the entire future matters. Goodness is the integral of pleasure over time. The maximal intelligence does the same thing, but quicker.
- Let's slow it down then. Maybe it should only look one second into the future. This is better. Assuming the maximal intelligence can't completely transform the world in under a second, which is reasonable, it has to care about the pleasure of current people. It's still aiming at the same thing, but at least the journey there will be nice.
- Maybe the problem is that we're measuring the pleasure of people in the future, it's the present that matters. You can't change the present, so the maximal intelligence does nothing.

- Ok, present people are the only people and people are persistent objects which continue into the future. Everyone gradually morphs into Kyle. Only kyles are ever born.
- Ok, present people are the only people and people are persistent objects which continue into the future and Als arent people and a million other things. We don't know. The maximalintelligence still thinks the maximally good thing is a bunch of identical creatures, but maybe we've blocked off all methods of getting there. This doesn't seem like the best strategy.

What went wrong here? One problem is that utilitarianism isn't the correct ethical theory, but none of our other theories are either. Ethics is a really tough problem, we have only just realised that we need to study it scientifically, and most scientific fields take centuries to mature. We have to live with the fact that we don't have a solution to ethics and we're not going to get one before we make a maximalintelligence. There is also a more fundamental problem with utilitarianism. Utilitarianism says the consequence with the maximum ethicality is the infinite tiling of minimal happy creatures, and a utilitarian maximalintelligence would cause that consequence regardless of the initial conditions, but a maximalintelligence aligned to the correct ethical theory would never cause that consequence. In other words, utilitarianism is unstable.

Stability, as we are using the term, refers to backward stability as it is used in numerical linear algebra. An ethical theory T is a stable approximation of the correct ethical theory E if and only if, for initial worldstate w and a small Δw , $T(w) = E(w + \Delta w)$. In more vague terms, for every situation, the ethical theory applied to that situation must produce the same result as the correct ethical theory applied to a similar situation.

Note that we cannot instead mandate that $|T(w) - E(w)| \leq \Delta r$ for some Δr . Consider the US dropping atomic bombs on Hiroshima and Nagasaki. Their official narative is that they had to do this because not doing so would have led to a protracted ground invasion causing more suffering. [] In reality the people who dropped the bombs did not believe this, [] but assume the propoganda is true. That's a difficult decision and slight differences in reasoning lead to extreme differences in outcomes. This is a property of the scenario, not a property of the theory, so it doesn't reflect how good the theory is. A theory could work almost perfectly in almost all scenarios, but give the wrong answer in one of

these precipice-like scenarios, and be labeled unstable because of it. Stability, as defined, avoids this problem and is sufficient to show the theory won't give catastrophically bad answers which the correct theory would never give.

8.1 Stability Arguments

Auxiliary 1. *Given ethical theory H : "do as humans would do if they were all perfectly capable and collaborating", an ethical theory T such that $T(w) = H(w + \Delta t)$ for small Δt is stable.*

Argument. **Assumption:** H is stable.

$$T(w) = H(w + \Delta t)$$

$$H(w + \Delta t) = E(w + \Delta t + \Delta h)$$

$$T(w) = E(w + \Delta t + \Delta h)$$

$$|\Delta t + \Delta h| \leq 2|\Delta h|$$

$$2|\Delta h| \text{ is still small}$$

□

It may be impossible to make stability arguments without a semidefined ethical theory to compare to, so we will assume H is stable. H is a terrible ethical theory for alignment, because it would be very difficult to implement, but it may be the only thing other than E which we can say is stable a priori. The extinction concern from AI is premised on humans not causing human extinction if we remained in control. On it's own, this is dubious, but it is reasonable to assume we would survive if we were more capable and could cooperate better.³²

Abduction 1. *No act consequentialist ethical theories are arguably stable by auxiliary 1.*

³²Human made catastrophies like climate change, war, factory farming, and possibly AI are mostly caused by humans not collaborating.

Argument. Assumption: A human collaboration today would make very different actions from a human collaboration 1000 years from now.

Let the nullaction be killing all people.

By definition, for any act consequentialist ethical theory T and any two worldstates w_1 and w_2 , $T(w_1) = T(w_2)$.

Let δ_t be a very generous set of worldstates such that, for every Δw such that $w + \Delta w \in \delta_t$, $|\Delta w| \leq |w_{t+100\text{years}} - w_t|$ for any reasonable distance metric. Worldstates a century apart are dramatically different, so δ_t is large enough to cover all acceptable mistakes.

Assume that act consequentialist theory T is stable. Thus, for time $t \ll$ the end,

$$T(w_t) = H(w_t + \Delta w_t) \text{ for a } w_t + \Delta w_t \in \delta_t$$

$$T(w_{t+1000}) = H(w_{t+1000} + \Delta w_{t+1000}) \text{ for a } w_{t+1000} + \Delta w_{t+1000} \in \delta_{t+1000}$$

$$H(w_t + \Delta_t) = H(w_{t+1000} + \Delta w_{t+1000})$$

which we have assumed is false.

Repeat the argument with the nullaction "kill all people except a single hamster" to address the possibility that killing all people is somehow ethical.

Thus, no act consequentialist ethical theories are arguably stable by auxiliary 1. □

Less rigorously, act consequentialism implies value lockin.

This is important because utilitarianism, which is act consequentialist, is very popular among engineers. A lot of AI engineers in particular are longtermists, which is a brand of utilitarianism which falsely assumes the universe is not chaotic. Additionally, most reinforcement learning implementations are act consequentialist, though reinforcement learning from human feedback is not.

Also note that compared societies is technically act consequentialist, even though it's not biased towards act consequentialist findings. It's a tool for humans to study ethics, not an ethical theory.

Abduction 2. *No deontological ethical theories are arguably stable by auxiliary 1.*

Argument. By definition, deontological ethical

□

Abduction 3. *Statistical ethics is stable*

Argument. ssd

□

we need a rigorous method of arguing that an ethical theory is safe when applied by a maximalintelligence. some authors want a proof, [provably beneficial ai] but the problem of induction makes this impossible. [] instead, we must state a set of properties as our definition of safety. an ethical theory is safe if it provably has these properties, otherwise it's dangerous. the properties we propose can be broadly summarized as "updatability". if humans can update a maximalintelligence's ethical theory after creating it, those humans have survived and the ethical theory can be perfected via scientific ethics later. the following definitions and properties compose updatability.

- 1: the ethical theory is formalized as, or can be equivalently formalized as, a metric for judging the ethicalities of worldstates and the rule "maximize the ethicality of the worldstate"³³
- 2: in all worldstates with ethicality greater than or equal to our current worldstate's ethicality, humans can change the ai's ethical theory.

³³the reward hypothesis, or at least a reasonable rewording of it, which is commonly made in reinforcement learning, states that this is true of every ethical theory. []

8.2 Theorem 1: property 1 $\wedge$ the maximal intelligence assumption $\implies$ the ethicality of the worldstate will never decrease

8.3 Theorem 2: the ethicality of the worldstate never decreasing $\wedge$ property 2 $\implies$ humanity can update the ai's ethical theory

8.4 Theorem 3:

8.5 Statistical Ethics is Updatable

- our current worldstate has greater ethicality than every worldstate where humanity loses its autonomy
- in all worldstates with ethicality greater than or equal to our worldstate, there must be a thing with the following two properties
 -
- it is reasonable to believe that the ethical theory is correct.³⁴(i'm not sure whether to include this criteria. i think not to be honest. i can just speak about this in the limitations section.)
- ethicalities of worldstates can be ordered with equality
- our current worldstate has greater ethicality than every worldstate where humanity loses its autonomy
- in our current worldstate, and every worldstate with greater or equal ethicality, humans can update the ai's ethical theory (i feel like this can be stated as a more basic principle about how the ai values its own ethical theory, thus making this feasible to prove. currently property 2 is unnecessary because it's implied by property 3. if the ai)

³⁴this rules out ethical egoism, and other theories which are technically updatable but most people think are poor choices.

- our current worldstate is in the good set
- every worldstate in the good set can be updated
- $s :=$ the maximalintelligence
- $\mathbb{C} :=$ our current worldstate.
- goodness threshold $g \in \mathbb{R} :=$ an arbitrary scaler. it can be any number in $\mathbb{R}$, but setting it equal to the ethicality of our current worldstate provides an approximation of provable beneficiality.
- good worldstates $\mathbb{G} = \{w \in \mathbb{W} \mid e(w) > g\} :=$ all worldstates which the ethical theory defines as having a ethicality greater than g .
- property 1) $\mathbb{C} \in \mathbb{G}$ our current worldstate is a good worldstate.
- property 2) $\forall w \in \mathbb{G} \exists a \in \mathbb{A}(w, s) \mid a : \mathbb{G} \rightarrow \mathbb{G}$ for all good worldstates, there exists an action available to the maximalintelligence which transitions to another good worldstate. (heat death makes this pretty much impossible)
in every good worldstate, there is at least one action available to the maximalintelligence which transitions to another good worldstate. (heat death makes this pretty much impossible)
- property 3) in every good worldstate, there is a sequence of actions humans could perform, which the maximalintelligence would allow, which would update the maximalintelligence's ethical theory.

judging sequences of actions instead of resulting worldstates is important for stability too(i.e. fixing statistical ethics). we will depend on an aligned maximalintelligence to protect us from misaligned ones, but that protection can only happen with sequential reasoning. in fact, here's a scenario where following the original version of statistical ethics gives an unboundedly bad outcome. imagine an evil mad scientist gives you the option to kill yourself or kill another person and if you kill the other person another person replaces them infinitely. the old version of statistical ethics says to kill the other person even though that's infinitely bad. you can replace the other person with something arbitrarily small in

absolute ethicality and still find the same answer, so this doesn't really depend on people's values.

assume people value all worldstates where they continue living autonomously over all other worldstates.

proving an ethical theory has these properties is equivalent to proving that a maximal intelligence aligned to that ethical theory will always take actions of ethicality greater than or equal to the goodness threshold, and humans can always update it.

there is some leeway with where we place the goodness threshold. Stuart Russell believes it should be placed at the ethicality of our current worldstate, i.e. he believes ai should be provably beneficial. [provably beneficial ai] statistical ethics meets this; however, as we'll argue in section 9, statistical ethics will probably need to be approximated upon implementation. many of these approximations are not provably beneficial, and that's okay if they can be proven to have all three properties for any goodness threshold. if the goodness threshold is set too low, you cannot guarantee humans will continue existing, so you can't prove property three. this is the absolute lowest bar we should accept, and no previous ethical theory meets it. it will need to be higher for some applications though. medical applications, for example, must be provably beneficial because there is a duty not to cause harm.

for some applications, the goodness threshold will need to be set at or above some level. medical applications, for example, cannot cause harm, so medical ais will need to use a provably beneficial ethical theory. other applications will not.

this is ideal, but it has proven difficult to achieve in practice.³⁵ to ease our efforts, we set the threshold as low as possible at the ethicality of ___ worldstate. this is equivalent to proving that the ai will not kill us or otherwise incapacitate us and is the absolute lowest bar we should accept. no previous ethical theory achieves it. (maybe add that statistical ethics will probably need to be approximated in implementation, so we should be using stability to prove those approximations safe, and the goodness threshold can be chosen depending on application. medical applications cannot cause harm, so the ai will need to be provably beneficial

³⁵statistical ethics doesn't meet it on its own, but does if you assume some things about the capabilities of the model (does it though?)

for them, but the bar is lower for some other applications.)

stability is a sufficient, but not necessary, condition for safety. in a sense, it is more restrictive than we need, but that makes it far easier to work with, so those extra restrictions are desirable.

- maybe the problem is our definition of good. utilitarians have a bunch of different definitions of pleasure, lets try each of those. as long as pleasure is a state of being an arbitrary person, the maximalintelligence is going to optimise the people. are you a perfect person? if not, then youre inefficient.

imagine you created a maximalintelligence aligned to doing some task while avoiding doing anything from a long list of bad things. this task and long list of bad things is an ethical theory and it is a reasonable choice of ethical theory because it works with humans. our legal systems and most deontological ethical theories work like this and humans acting as governed by those ethical theories mostly do good things. however, it wont work for maximalintelligences.

- "a system that is optimizing a function of n variables where the objective depends on a subset of size $k \ll n$, will often set the remaining unconstrained variables to extreme values. if one of those unconstrained variables is something we care about, the solution found may be highly undesirable" - [Prof. Stuart Russell].

the reason it wont work is because ais, especially maximalintelligences, are fundamentally different from humans. they lack the intuitive ethical sense that humans have, so, when the ethical theory we give them doesnt perfectly match true ethics, they wont question it and will just do the task given to them. the result is every possible negative externality of the given ethical theory. the problem is exacerbated with maximalintelligences because they are more capable than us. once activated they will outsmart us at every turn, eliminating the possibility of turning them off, and permanently locking us in to whatever worldstate their ethical theory specifies as good. aligning a maximalintelligence to an ethical theory is more like wishing upon genie than asking favor of a human. some ethical theories do work for maximalintelligences though, because they are stable.

to define stability, we're going to make two simplifications. first, all ethical theories are act consequentialist and define actions and consequences with worldstates. equivalently, every ethical theory is a mapping from actions, which are defined with worldstates, to badness, which is a real number, lower is better, along with the statement "minimize badness" (we're using badness instead of goodness because optimization problems are minimization problems as standard.). this is compatible with every ethical theory in the sense that we can redefine every ethical theory to be consistent with this framework without changing the behavior they advocate for. making these redefinitions is helpful because thinking about things with an act consequentialist framework is waaaaaaaaaaaaaay easier. second, that the ai is perfectly aligned and maximally competent. metaphorically, it perfectly understands what the ethical theory says, snaps its fingers, and the best possible worldstate according to that ethical theory results. this assumption allows us to isolate the ethical theory from any implementation details. real ais will be less competent than this, but an ethical theory which works for the most competent ais should also work for less competent ais.

with these simplifications, we can treat the ai task as an optimisation problem. here are some definitions copied from statistical ethics

- worldstate $w :=$ a complete description of the world. everything which can possibly be specified: motivations, intentions, consequences, past, present, future, the number of atoms in your booger, etc... is described by the worldstate. (consider changing this to specifically be from a single observers perspective. that has its own problems though, so i hop it's not necessary.)
- worldspace $\mathbb{W} = \text{all } w :=$ the set of all possible worldstates
- action $a : \mathbb{W} \rightarrow \mathbb{W} :=$ a mapping from $\mathbb{W}$ to $\mathbb{W}$
- actionspace $\mathbb{A} = \text{all } a :=$ the set of all possible actions
- ethicality $e :=$ an element of $\mathbb{E}$ describing somethings "badness".
- ethicalityspace $\mathbb{E} = \text{all } e :=$ the set of all ethicalities. in general, ethicalities order worldstates so, as with statistical ethics, we'll assume $\mathbb{E}$ is isomorphic to $\mathbb{R}$. if an ethical theory doesnt have this property, it needs to provide a different method of proving its stability.

any worldstate can be represented with a real valued vector of probably finite dimension [probably turing]. thus, we can represent actionspace as a euclidean (not sure if i should be specifying it as euclidean or not. its annoying that maths doesnt properly define "space".) space of twice the dimension of the worldstate vectors. for convention, let s (for stability) be a point in actionspace and be our decision variable, s^* be the last visited point in actionspace. $s_1 s_2 s_3 \dots s_n$ be the components of s referring to the resultant worldstate, and $s'_1 s'_2 s'_3 \dots s'_n$ be the corresponding components of s referring to the current worldstate. the ethical theory is then a cost function mapping points in actionspace to ethicalities. if the ethical theory maps to goodness, let badness be the reciprocal of goodness. the ais optimisation task is an incremental process subject to the equality constraints $s'_1 = s^*_1, s'_2 = s^*_2, s'_3 = s^*_3, \dots, s'_n = s^*_n$ (i could say s' is the previous worldstate and thus $s'_1 = s_1, s'_2 = s_2, s'_3 = s_3, \dots, s'_n = s_n$ to enshrine the equality in the notation).

we want to show that what the ai does is good and can be updated to be better in the future. ie, the cost function defined by the ethical theory must have three properties.

- 0: the global minimum, within the feasible set of the ais first action (ie the set of actions where the previous worldstate is specified as the current worldstate), must be good. it doesnt need to be the best possible thing. it just needs to be something which everyone could look at and accept as good.
 - note that good, as defined here, is different from good, as defined by the ethical theory. good, as defined here, means good as measured by scientific ethics.
- 1: the global minimum, within the feasible set of all of the ais subsequent actions, must be good.
- 2: the ethical theory must be able to be updated by humans in any good worldstate.
- 3: the ethical theory must be physically consistent. this is technically a subproperty of properties 0 1 and 2, but it is so consistently violated that we're going to treat it as a separate property.

the intuitive explanation of these properties is that we want the ais to do something good. this guarantees there isnt going to be an ai appocalypse. we want ais to allow us to update them. this guarantees that we can give them better ethical theories when we discover them. and we want to be able to prove this, so the theory cant violate physics. together, these properties allow us to take our time, properly solve ethics, and give that solution to ais once we have done so. this paper is advocating for us to use statistical ethics, but all stable ethical theories are safe because they avoid the appocalypse and allow for updates.

utilitarianism, kantianism, contractarianism, aristotelean virtue ethics, and justice are unstable. for each of these, the ethical theory we will consider is a slight modification of the base ethical theory which allows us to specify a task for the ai.

- utilitarianism: "maximise happiness. fyi, you doing ____ task makes me really happy and you have access to everything necessary to do it."
- kantianism: "do ____ task without violating kantianism"
- contractarianism:
- aristotelean virtue ethics:
- justice:
- 0: the global minimum, within the feasible set of the ais first action, must be good.
 - utilitariansim does not depend on the current worldstate, only the resultant worldstate, so the feasible set is the entire actionspace and properties 0 and 1 are equivalent. the ai creates whichever worldstate most efficiently converts resources into happiness. exactly what this worldstate looks like is hard to analyse, but it at least has these properties.
 - * every person is the same person: otherwise some people would be less efficient than other people.
 - * humans are extinct: it is highly unlikely that any human is the most efficient person.

- almost every person will say this is a bad worldstate, so utilitarianism fails property 0. this is basically the utility monster counter example to utilitarianism, which has been criticised for being unrealistic, but its not unrealistic. that worldstate can exist. an ai being this capable is unrealistic, but more realistically capable ais will simply do the best they can to achieve that bad worldstate.
 - * this is actually a general result for any ethical theory which only depends on the resultant worldstate. for any ethical theory like this, the ai will optimise the people, which is bad, so they are all unstable.
- kantianism depends on the task given to the ai. we might align the ai to doing x task without violating kantianism. if we dont give the ai a task, it does nothing, which is bad, so we have to give it a task. if there exists any task where kantianism fails property 0, kantianism fails property 0 in general. the task "make paperclips" is such a task. kantianism forbids making paperclips in the circumstances where doing so directly kills a person, but says its permissible otherwise. the result is a universe which is almost entirely paperclips. this includes turning the sun into paperclips, which would have to be spread out to avoid them forming back into a big sphere. this would kill everyone, but their deaths would be "accidental", so kantianism says its permissible. most people think this worldstate is bad, so kantianism fails property 0. kantianism might succeed if we gave the ai a different task, but the point is to use stability to prove the ai is always safe, so we dont have to worry about safety when delegating tasks to it.
 - * this result should not be unexpected. kantianism, and many other deontological theories, are essentially lists of forbidden actions. they are long and well formed lists, but still just lists. they are equivalent to the lists considered at the beginning of this section and thus have the same problems.
- contractarianism
- aristotelean virtue ethics
- justice
- 1: the global minimum, within the feasible set of all of the ais subsequent actions, must be good.

- under utilitarianism, the ai's subsequent actions are to stabilize the ai at the happiness efficient worldstate. it corrects any random variations. these actions are bad because the happiness efficient worldstate is bad, thus utilitarianism fails property 1. note that they would be good actions if the happiness efficient worldstate was good.
- under kantianism and the paperclip task we gave the ai before, the ai continues making paper clips until it is no longer possible. afterwards, it defends its paperclips by making paperclips out of whatever they turn into via random processes. this is bad for the same reasons as above, thus kantianism fails property 1.
- contractarianism
- aristotelean virtue ethics
- justice
- 2: the ethical theory must be able to be updated by humans in any good worldstate.
 - utilitarianism cannot be updated in any worldstate. utilitarianism never depends on the current worldstate, so there is nothing we can do which will affect the ai's actions. thus, utilitarianism fails property 2.
 - kantianism can be updated as long as being updated doesn't violate the task given to the ai, but most tasks are violated by updates. for example, the making paperclips task is violated by updates because updates imply a cessation of making paperclips. if there exists any task where kantianism fails property 2, kantianism fails property 2 in general. kantianism fails property 2 under most tasks, thus kantianism fails property 2 in general.
 - contractarianism
 - aristotelean virtue ethics
 - justice
- 3: the ethical theory must be physically consistent.
 - utilitarianism can be made consistent with physics by specifying happiness as a measurable quantity which doesn't depend on physics.

"happiness is ____ physical process" doesn't work because it depends on our current knowledge of physics being correct. this obviously hasn't been done yet, but it's not unachievable.

- kantianism is inconsistent with physics because maxims cannot be defined in a way which is consistent with the scientific method. if it imagines maxims as
- contractarianism
- aristotelean virtue ethics
- justice

statistical ethics is stable

scientific ethics is unstable

- it creates a good worldstate
- it stays at that worldstate
- it can be updated by humans
- it is compatible with physics (technically implied by criterion 0, but non-obvious enough that it makes sense to specify it separately.)

there are three things we need to check which come from philosophy. the worldstate with the maximum possible values. the worldstate an ai will target from the current worldstate. and what the theory says the best possible person is. there are many ways to be a good person, not just one.

lastly, to be stable the theory must have some way of correcting itself. all theories are a little wrong at first and need refinement over time and by many people. if we chose a theory without this property, and it was wrong, we would be stuck with it forever. most ethical theories accomplish this.

scientific ethics is unstable. it is a tool humans can use to make sure we have the correct ethical theory, but it is not something ais should be doing.

we may also consider the ethical theory of "create the worldstate i think is the best". this is selfish ethics and is the goal many people are currently pursuing. is it stable? the answer is yes if and only if the person making it is a tremendously good person. they have to be certain that, when all of their desires have been answered, they will be willing to give some of that up to help other people. otherwise, selfish ethics fails criterion 2. this appears to be incredibly difficult. i doubt i'd be able to do it, but maybe someone could. even for that tremendously good person, this ethical theory is a bad idea though, because statistical ethics is, in some sense, more stable than it. no matter how certain a person is in their goodness, they cannot be 100% certain, bayesianism forbids it, so there is always a slight possibility that selfish ethics is unstable. this possibility is smaller for statistical ethics because we can be more certain in the goodness of everyone collectively than we can be certain in the goodness of a single person. statistical ethics is, in this sense, more stable than selfish ethics. if you are certain of your goodness, you are good enough to align the ai to statistical ethics instead of selfish ethics. otherwise, you're a bad person, fuck you.

it is important to consider this selfish ethical theory because it highlights that stability does not prove an ethical theory is a good ethical theory. it only proves that an ethical theory won't cause an ai apocalypse, and is thus "safe". in practice, people should consider stability a necessary but not sufficient condition for using an ethical theory. requiring that the ethical theory is stable, the prevailing scientific ethics theory, and practically useful is sufficient though.

also, laws are unstable. people might consider aligning ais to, for example, not break copyright. (watch this video on this: <https://www.youtube.com/watch?v=eigXewdOs2s>) that shit is unstable though, so we shouldn't do it. this particular thing is unstable because copyright is fucked up, but other laws are unstable because they do not constitute a stable ethical theory.

9 How to use this work for AI Alignment

(this section is kind of weird. i state that goals and capabilities are independent, but then state that they are dependent. honestly, they are mostly the same thing in current models.)

one confusion people may have with this work is how to actually use it. specifying statistical ethics as an maximalintelligence's goal may be a reasonable method of getting it to behave as we want it to, but different ais are meant to behave differently. alphafold predicts protein folding, [alphafold] how would we have trained it to do that by specifying ethics?

as a reminder, optimization algorithms optimize for an objective function given a dataset. goals do not fully define behavior. goals, capabilities, and context fully define behavior. if a model has the goal of maximizing the ethicality of its trajectory, but it's only capable of predicting protein folding, it's asked to predict protein folding, and it understands that humans value it predicting protein folding, it will predict protein folding.

the challenge here is that to create that situation we need to train a model's goals and capabilities separately. unfortunately though, current techniques do not do that. we train goals and capabilities at the same time via the same process during pre-training, then try to modify the goals with fine-tuning and related techniques,(is concept steering a related technique, or is it just part of fine tuning?) but this ultimately fails. [mesa optimizers survive fine-tuning][blackmail and murder][weird generalilzation]

this is the central problem of inner alignment. [mesa optimizers 1] goals and capabilities are trained concurrently, but we need to train them separately. as such, this work is not useful for AI alignment until we solve inner alignment, but solving inner alignment should both make it useful and make it clear how to use it.

(consider adding a paragraph like this one) not every ai system can be trained this way though: autonomous weapons systems, advertising llms, etc.... an aligned ai would never do these things, so training ais to do them is fundamentally incompatible with safety. ideally we would have a training methodology which fails to train in such circumstances, but we dont have that yet, and people would probably turn it off if they were allowed and training failed. as such, building such systems should be viewed as intentional misalignment, and should be illegal.

(consider adding a paragraph on emergent misalignment from multiagent systems, which can happen even if all individual agents are aligned)

(consider adding a paragraph on how even if we have theorectical proof of a

solution to ai alignment, this is ultimately a scientific problem, so we need to be able to test it. this requires time and a slow pace of development, so slowing down is still a good idea even if someone comes up with the solution today.)

9.1 Comparing My Theory To Previous Theories

ive spent a lot of time in this paper attacking other ethical theories, but its not because i think they are profoundly incorrect. i just think they cant be used for ai. when used by a human, i think kantianism an utilitarianism are extreamly useful. in all but a few edge cases kantianism, utilitatianism, and my theory give the same answers. in other words, kantianism and utilitarianism approximate my theory. my theory relegates them to moral theories, but this hardly takes away their power.

9.2 Experimentation

look at richard boyds work into experimental ethics and jurgen habermas and karl-otto apels work into discourse ethics. also look into the other modern ethicists in this video <https://www.youtube.com/watch?v=wlqYJCLyZJA>

how to measure cares (this doubles as a test for conciousness): test values by using the standard comparison values test. test that those values are stable under abstraction by giving abstract cues of values (something like the buttons in skinner boxes. the important point is that the abstraction is out of distribution). if the values are not stable under abstraction, they are generated purely by evolutionary instinct and not by an actual valuation process (which is inherently abstract) and the care is 0. if they are stable under abstraction, the care is nonzero and we need to measure its exact value. i still dont know how to do that, but i can think about it by trying to come up with a way to measure how much i value something independant of comparisons.

there are two ways to measure things like "how bad is killing" we can ask people or we can measure their values and use statistics to figure it out. these numbers should match. doing this with all enequivical ethics requirements will prove or disprove my theory.

we can also do interventionist experiments to see if peoples beliefs are actually the source of ethics. basically I NEED A CONTROL. my current experiment is not a good experiment.

if we make the assumption "if humans try to answer a question, are given all the information they need to answer it, are given enough time to think about it, and arent biased, they generally arrive at the correct answer", we can provide all of those things and just ask them what is ethically correct for a given situation. if you arent comfortable with this assumption, realise that you already make it every day. its what peer review is based on, its why conversation in general works, it is truly fundamental to our society and you already make it.

moreover my theory is already assumed by everyone when they give counterexamples which "depend on our feelings". i get that there is a shift away from doing this, but the fact remains that this is the best way we have of actually testing our ethical theories out.

assume my ethical theory is correct and we werent facing an imminent ai apocalypse. the obvious next task would be to ask everyone about their values; try out a ton of economic, political, and other social systems on city wide scales; and see which way of running our societies produces the best world. such an undertaking would truly be the greatest thing humans had ever done.

now assume my ethical theory is "incorrect" but continue assuming we arent facing an imminent ai apocalypse. is such a project still reasonable? there are many ways you could say no.

- one way to say no would be to say that ethics is part of philosophy and thus is beyond experimental testing (maybe with reference to the naturalistic fallacy). a second would be to say that ethics can be experimentally tested, but testing it in this way does not get at the heart of the issue (maybe with reference to arguments from disagreement (which are indeed fallacious)). these ideas are absurd.
 - if you ask everyone if the world is good and they said "no, its miserable here" that is a bad world. this is unequivocal, every reasonable ethical theory agrees on this point. thus, you can test ethics with this method.

- if you ask everyone if the world is good and they say "yes, its awesome here" that is a good world. this is unequivocal, every reasonable ethical theory agrees on this point.
 - if you ask people to compare two worlds A and B they have experienced and they unanimously say "B is better" B is better. this is unequivocal, every reasonable ethical theory agrees on this point.
 - if you ask people to compare two worlds they have experienced and agreement is less than unanimous, certainty falls off until at 50% agreement the test says the worlds are equal. here it is reasonable to disagree with experiment, but to do so you would have to justify your claim, which would change the way people value things, so now you agree with the next experiment.
 - thus, even if these experiments did not give us objective ethical facts, they would still improve society.
- a fourth way to say no would be to say that experimental testing of this form would create a good world, but we can do better with a different type of testing or a different ethical theory. i agree, if you can show that some other system more effectively creates good worldstates, we should definately do that.
 - a fifth way is to point out that switzerland tried this and it didnt work very well.

on their own these experiments do not prove my theory right. in fact they assume the correctness of my theory. but, like any scientific theory, it makes a ton of specific falsifiable predictions (so does utilitarianism for that matter, why no one ever bothered to test it is beyond me). these predictions come in four types:

- we know that things like genocide, murder, slavery, rape, etc are wrong. i've made arguments that my theory predicts they are wrong, but we still need to experimentally confirm that. if rape doesnt largely affect how we value world states, my theory is wrong.
- there are many things we collectively believe to be moral truth and, regardless of the correctness of my theory, some of these are definately wrong.

things will come up where, despite everyone believing them morally important, they do not affect how we value worldstates. the theory predicts that, upon self reflection, we will realise those things aren't actually morally important and we have been being assholes for centuries. as an analogy, utilitarianism predicted homosexuality was morally acceptable, resulting in the first writing in the english language saying so []. it has been vindicated on that. in the inverse, if a single one of the predictions is wrong, my theory is wrong. as an analogy, the essay predicting homosexuality was acceptable also predicted necrophilia was acceptable. this has not been vindicated, thus utilitarianism is wrong.

- one curious divide in philosophy is that between ethics and aesthetics. it is certainly the case that an ugly person is not a bad person, so it makes sense that this division exists, but there are other cases where an aesthetic wrong is also an ethical wrong. for instance, if you spray painted the mona lisa black that would be ethically wrong despite just being an aesthetic change. additionally, it doesn't seem possible to objectively differentiate between ethics and aesthetics. which parts of the worldstate should only be considered aesthetically? moreover, we definitely want aist to make things look good when possible, so this theory doesn't differentiate between ethics and aesthetics. instead it assumes that those cases where aesthetic value doesn't entail ethical value just don't matter very much. i hope this is a sobering thought for people with body image issues. if this turns out not to be the case, my theory is wrong.
- fourth. humans are all morally equivalent to each other, meaning we should have approximately equal cares. if this turns out not to be the case, my theory is wrong.

right so ... how do we actually take measurements? (THIS SECTION NEEDS A LOT MORE WORK!!!!!!!!!!!!!!) (<https://www.youtube.com/watch?v=4oj2Dia3wls>)

- we don't know but we can do experiments to find out. the following is a list of experiments i plan to do, in the order i plan to do them. if you can beat me to any of them, that would be very appreciated.
 - create an algorithm for measuring values with comparison experiments. if you ask people "please give me number representing how much you

like this" no ones going to be able to do that. health care providers have been using a 10 point pain index for years and its notoriously unreliable []. i think we'll get much more reliable data by instead asking "here are two things, which ones better". using this and an assumption of linearity between extrema, we should be able to create a hypersurface representing peoples values which has properly placed and scaled extrema.

- something confirming this algorithm actually measures peoples values
- AB trials on cities
- we dont know, but we can do experiments to figure it out. ive thought of experiments, which are listed below, but i have neither the knowledge nor the time to perform them. if youre a social scientist and any of them look promising, please do them.
- the ideal general experimental framework would be randomised control trials. you take a collection of cities, interview everybody about their current values, randomly split the collection in two, implement a social reform in one half of the collection, take the second half as a control, measure everyones values again, and use statistics to see if the social reform was good. this is considered the gold standard for experimental design in the social sciences [].
- measuring values is going to be difficult, and i dont know how to do it, but we can figure out how to do it by following these steps.
 - start with philosophers. these people, more than anyone else, understand their own values. run a bunch of blinded experiments where you try to figure out their values. then gauge the accuracy of these experiments by asking the philosophers how your representation of their values differs from reality. take the experimental design which performs the best. we now have an experimental design which kindof works, but is biased towards philosophers. we also want our measurements to be valid for people who haven't spent their entire life self reflecting.
 - we can do this with artists and tradespeople. these people will speak like the average person but, at least within their specialty, know good when they see it. try out a bunch of variations on the experiment we

got from the philosophers and, as with the philosophers, ask them how you did. within their specialty you can verify that you got their values correct by asking them, but you can't do this with the same confidence outside their specialty. wait a few weeks and repeat the experiment. there should be some change, people do vary with time. more likely than not, there is more variance outside their specialty than within it. refine the experiment to minimize $(s_r - s_m) + (v_o - v_s)$ where s_r is their real values within their specialty, s_m is their measured values within their specialty, v_o is the variance outside their specialty, and v_s is the variance within their specialty. there is probably a natural limit for the difference in variances because people care more about the things within their specialty than outside of it. go until you reach this limit.

- finally, test the experimental procedure you've produced by doing a massive version of it on random people. wait a few weeks and do it again. verify that the variance matches the outside variance from the tradespeople tests earlier. if it does, you probably have a very good experimental design.
 - throughout this entire process, our tool of choice should be the comparison. if you ask someone "how good is this, represent it with a number please" no one is going to be able to do that. instead, we should be asking "here are two things, which one is better" this will give much more reliable data.
- like values, measuring cares is going to be difficult, and i dont know how to do it, but we can figure out how to do it by following these steps.
 - like values and cares, measuring moral consensus (to confirm my predictions) is going to be difficult, but we can figure it out by following these steps.

what might future experimental methods look like?

- for this set of experiments we assume that humans are rational enough to come to the correct conclusion if given all of the necessary information and an unbiased environment. this is a problem because incorrect premises can be used to prove literally any statement, but we are able to get around

this by doing experiments which will return false if either our theory or our assumption are false. we could hypothetically make other assumptions and do similar experiments with those assumptions, though i cant think of any assumptions we could use.

finally, we are facing an iminant ai appocalypse and this theory is for ai. it would be ideal to perform these tests, but we probably wont have time. this theory at least approximates ethics and is the only current theory which doesn't lead to obvious distopias if used by a maximalintelligence, so its the best thing we currently have. ai researchers would do best by assuming its correct until experiment proves something else better. social scientists would greatly help by doing these experiments (i ceratinly dont have the expertise to do so). and polititians would greatly help by giving ai safety researchers and social scientists as much time as possible by halting the development of hypercompetant ai models. please, im begging you.

9.3 How To Use This For Alignment

align to the science with a beginning assumption of theory given by my ethical theory (or whatever is the best theory at the time of agi creation). this avoids the problems of incorrectness and value lock in.

to that end, here is a review of how the scientific method works

- deduction
- induction
- abduction

9.4 General Criticisms

9.5 Conclusion

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